Hourly Air-Quality Forecasting for Nashik Monitoring Stations with KNN Imputation and a CNN-LSTM Network: 2025 Backtest and 2026 Forecast

Academic Project Report

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Abstract

Hourly concentrations of seven pollutants (PM2.5, PM10, NO2, SO2, CO, O3 and NH3) from four continuous monitoring stations in Nashik, India (MIDC Ambad, Gangapur Road, Pandav Nagari and Hirwadi) were used to train one convolutional-LSTM (CNN-LSTM) network per station that predicts the next hour from the previous 24. Missing inputs were filled with a K-nearest-neighbours imputer fitted on training data only. The pipeline runs in two separate modes. The 2025 backtest trains on data up to 31 December 2024 and forecasts every hour of 2025, which is fully measured at all stations. The 2026 forecast trains on data up to 31 December 2025 and forecasts all 8,760 hours of 2026 at all four stations; measurements exist only for January-June and August 2026 at three stations and not at all at Pandav Nagari. Two kinds of prediction are scored separately on measured hours: a recursive forecast of the whole year from the forecast origin, which uses no later measurement, and a rolling one-step evaluation, in which the frozen model predicts each measured hour from the measured previous 24 hours. Pooled over stations, with quality-flagged sensor series excluded, the recursive forecast changed RMSE relative to an hour-of-day climatology by -111.6% (NH3) to -9.3% (O3) in 2025 and -89.5% (PM2.5) to -32.5% (CO) in 2026 (negative = worse than climatology): fed its own predictions for a year, the network is a worse year-ahead estimate than the hour-of-day climatology for every pollutant. The one-step predictions changed RMSE relative to persistence by -7.3% (NO2) to 7.9% (SO2) in 2025 and 3.3% (NH3) to 11.3% (NO2) in 2026: a small, mostly consistent gain when fresh measurements are available every hour. Daily CPCB AQI values were derived from the predicted concentrations; they are model estimates, not official AQI bulletins.

Keywords

air-quality forecasting; CNN-LSTM; KNN imputation; recursive forecasting; backtest; CPCB AQI; Nashik; chronological validation

1. Introduction

Short-term air-quality prediction supports public advisories and operational decisions such as traffic or industrial-emission controls. Continuous ambient monitoring stations report hourly concentrations, but their records contain gaps, instrument faults and changes in data availability, so any forecasting pipeline must handle missing data without using information from after the forecast origin.

Hybrid convolutional-recurrent networks are a common choice for such series: one-dimensional convolutions extract short local patterns, and Long Short-Term Memory (LSTM) layers model longer dependencies [1], [3], [4]. This project applies that design to four Nashik stations, produces an hourly forecast of 2026, and tests the same procedure on 2025, where the full year was measured.

2. Problem Statement

Using only the hourly observations of a station available before 1 January 2026, forecast the concentrations of seven pollutants for every hour of 2026, and measure the accuracy of the same procedure where the answer is known: on 2025, with data up to 31 December 2024, and on the hours of 2026 that have since been measured.

3. Objectives

- Document the dataset's schema, coverage and quality problems per station.
- Build reproducible cleaning and KNN imputation fitted without temporal leakage.
- Train one CNN-LSTM per station on data before each forecast origin.
- Export a continuous hourly 2026 forecast for all four stations, including hours and stations without measurements, without fabricating ground truth.
- Score recursive and one-step predictions separately with MSE, RMSE, MAE and R^2 in original units, against climatology, same-hour-last-year and persistence baselines.
- Derive daily CPCB AQI from the predicted concentrations, clearly separated from measured AQI.

4. Dataset Description

The dataset consists of four CSV files, one per station. Station identity is given only by the file name. All files share 25 columns: a Timestamp plus 24 measurements (criteria pollutants, volatile organic compounds and meteorological variables). The Hirwadi file has an extra Source_File column whose labels do not match the timestamps, so it was dropped and the timestamps were treated as authoritative. Hirwadi rows are stored out of order and were sorted. No file contains duplicate timestamps.

Table 1. Raw files as delivered.

Station	Rows	First timestamp	Last timestamp	Duplicate times	Sorted	Years present	Rows with no pollutant
MIDC Ambad	40152	2017-01-01	2026-08-31 23:00:00	0	True	2017, 2023, 2024, 2025, 2026	14377
Gangapur Road	81024	2017-01-01	2026-08-31	0	True	2017, 2018,	4945

			23:00:00			2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026	
Pandav Nagari	35064	2017-01-01	2025-12-31 23:00:00	0	True	2017, 2023, 2024, 2025	12761
Hirwadi	40152	2017-01-01	2026-08-31 23:00:00	0	False	2017, 2023, 2024, 2025, 2026	13873

Table 2. Coverage of each forecast horizon in the station files.

Station	Horizon	Rows in file (of 8,760)	Hours with a measurement	First measured	Last measured	Months with no rows
MIDC Ambad	2025	8760	8456	2025-01-01 00:00:00	2025-12-31 23:00:00	none
Gangapur Road	2025	8760	8511	2025-01-01 00:00:00	2025-12-31 23:00:00	none
Pandav Nagari	2025	8760	8435	2025-01-01 16:00:00	2025-12-31 23:00:00	none
Hirwadi	2025	8760	8550	2025-01-01 00:00:00	2025-12-31 23:00:00	none
MIDC Ambad	2026	5088	4739	2026-01-01	2026-08-31 23:00:00	Jul, Sep, Oct, Nov, Dec
Gangapur Road	2026	5088	4989	2026-01-01	2026-08-31 23:00:00	Jul, Sep, Oct, Nov, Dec
Pandav Nagari	2026	0	0	-	-	Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec
Hirwadi	2026	5088	4844	2026-01-01	2026-08-31 15:00:00	Jul, Sep, Oct, Nov, Dec

Three files begin with an empty 2017 block and resume in mid-2023; Gangapur Road has data from 2017 onwards. In 2026, every file that has 2026 rows covers 1 January-30 June and 1-31 August (5,088 of 8,760 hours); July and September-December are absent. Pandav Nagari's file ends on 31 December 2025. Its 2026 forecast is produced like the others but cannot be scored.

Seven pollutants are the prediction targets: the pollutants of the Indian National AQI present in the data (PM2.5, PM10, NO2, SO2, CO, O3, NH3). Meteorological variables were not used: temperature and humidity are absent at Gangapur Road before 2026, and wind speed shows level shifts consistent with sensor changes.

5. Data Preprocessing and KNN Imputation

5.1 Cleaning rules

- Timestamps parsed and sorted; a duplicate timestamp would keep its first row (none occur).
- Rows before a station's first pollutant observation removed (the empty 2017 blocks).
- Negative concentrations set to missing (none occur). Zero is a valid value and is kept, unless it repeats as a stuck reading.
- PM2.5 or PM10 of 999 or more set to missing: the analyser ceiling (21 values).
- Any value repeated unchanged for 12 or more consecutive hours set to missing as a stuck-sensor reading (13,921 values).

- No interpolation or resampling. Hours absent from a file are not filled, and no input window spans them.

5.2 Missing values

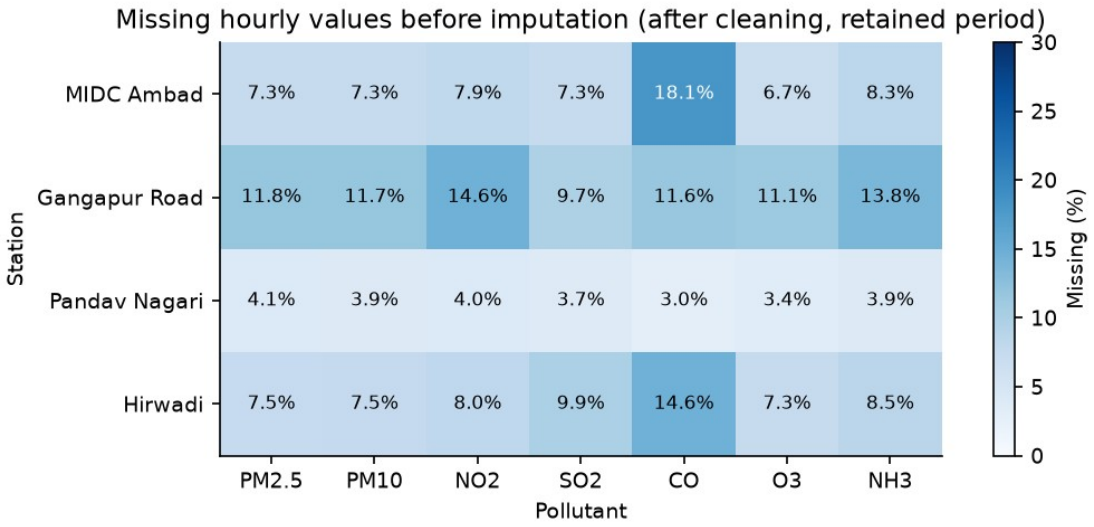


Figure 1. Missing hourly values per station and pollutant before imputation (all data up to the end of the 2026 horizon).

Table 3. Percentage of missing hourly values before imputation.

Station	PM2.5	PM10	NO2	SO2	CO	O3	NH3
MIDC Ambad	7.3	7.3	7.9	7.3	18.1	6.8	8.3
Gangapur Road	11.8	11.7	14.6	9.7	11.6	11.2	13.8
Pandav Nagari	4.1	3.9	4.0	3.7	3.0	3.4	3.9
Hirwadi	7.5	7.5	8.0	9.9	14.6	7.3	8.5

5.3 KNN imputation

A missing value is filled with the inverse-distance-weighted mean of that variable in the k = 5 most similar hours, using the NaN-Euclidean distance over the seven standardised pollutants of the same hour [2], [5], [6].

- One StandardScaler and one KNNImputer per station and mode, fitted on that station's training rows only, then applied unchanged to validation, test and forecast inputs. Donors are always training-period hours.
- Imputed values are model inputs only. Training targets are never imputed (the loss is masked) and every metric uses measured values only.
- Hours with every pollutant missing get the training mean; windows with more than half of their pollutant cells imputed are discarded.

Table 4. Missing pollutant cells before and after KNN imputation, 2026-forecast mode.

Station	Split	Missing cells before	Missing cells after

MIDC Ambad	train	9802	0
MIDC Ambad	val	1664	0
MIDC Ambad	test	5818	0
Gangapur Road	train	58622	0
Gangapur Road	val	1423	0
Gangapur Road	test	8310	0
Pandav Nagari	train	4504	0
Pandav Nagari	val	1451	0
Pandav Nagari	test	0	0
Hirwadi	train	11641	0
Hirwadi	val	1014	0
Hirwadi	test	5052	0

6. Exploratory Data Analysis

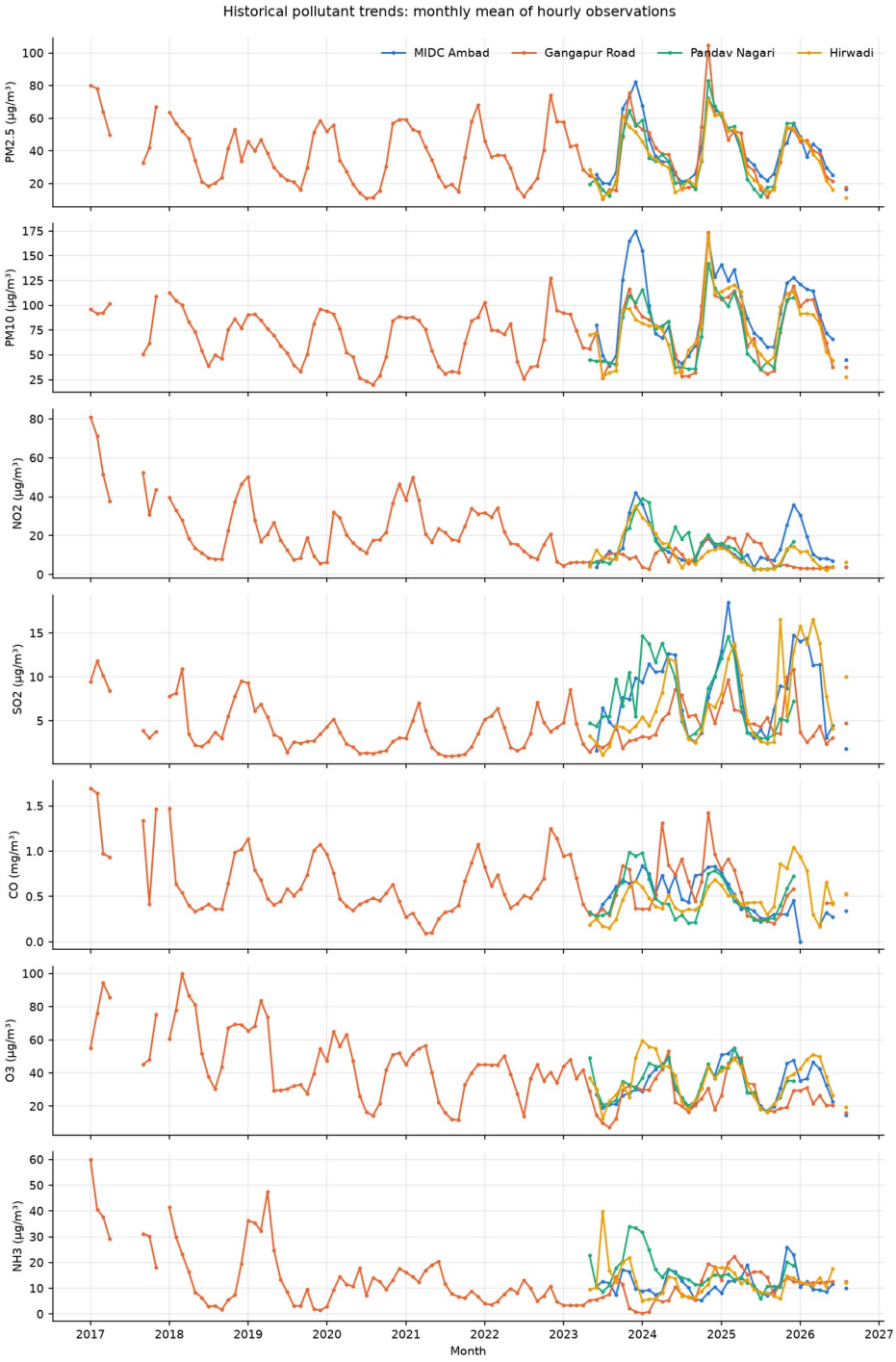


Figure 2. Monthly means of hourly observations per pollutant and station. Gaps are periods with no observations.

Particulate matter follows a seasonal cycle at all stations: low during the monsoon (June-September) and high in winter. This cycle is what a year-ahead forecast can realistically capture. Gangapur Road's NO2, CO and NH3 series change character in 2026 (few distinct values, near-constant levels), consistent with an instrument or calibration change.

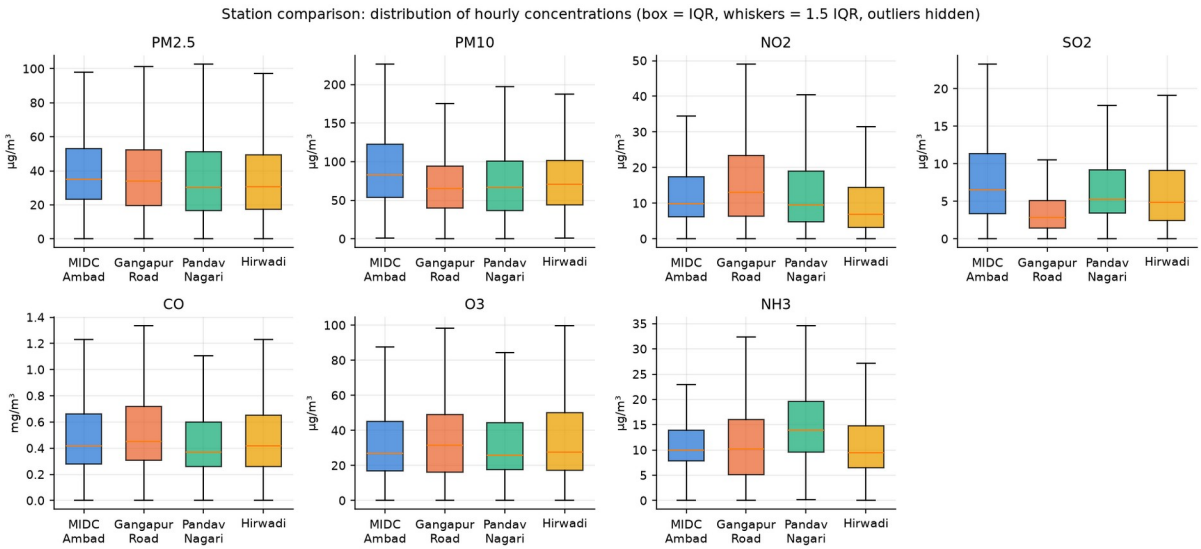


Figure 3. Distribution of hourly concentrations by station (box: interquartile range; whiskers: 1.5 IQR; outliers hidden).

7. CNN-LSTM Architecture

Two causal one-dimensional convolutions (64 filters, kernel 3) extract local temporal motifs from the 24-hour window; max-pooling halves the sequence; an LSTM [1] with 64 units summarises it; dropout [9] regularises; two dense layers output the seven pollutants. Inputs per hour: the seven standardised, KNN-imputed pollutants plus sine and cosine encodings of hour-of-day, day-of-week and month (13 features). Calendar encodings are known for any future hour.

Table 5. CNN-LSTM layers (50,247 trainable parameters per station model).

Layer	Output shape	Configuration	Parameters
Input	(24, 13)	24 hours x 13 features	0
Conv1D	(24, 64)	64 filters, kernel 3, causal padding, ReLU	2,560
Conv1D	(24, 64)	64 filters, kernel 3, causal padding, ReLU	12,352
MaxPooling1D	(12, 64)	pool size 2	0
LSTM	(64)	64 units, final hidden state	33,024
Dropout	(64)	rate 0.2	0
Dense	(32)	ReLU	2,080
Dense (output)	(7)	linear, one unit per pollutant	231

8. Methodology

8.1 Two modes

Table 6. Run configuration. Each mode trains its own models, scalers and imputers and writes to its own output folder.

Mode	Training cutoff	Horizon start	Horizon end	Horizon hours
2025 backtest	2024-12-31 23:00:00	2025-01-01 00:00:00	2025-12-31 23:00:00	8760
2026 forecast	2025-12-31 23:00:00	2026-01-01 00:00:00	2026-12-31 23:00:00	8760

Within each mode, validation is the six months before the forecast origin and training is everything earlier. Scaler, imputer, network weights, early stopping and the choice of target design use only data up to the training cutoff. Horizon rows are used for two things only: as ground truth, and as inputs to the one-step evaluation.

8.2 Two kinds of prediction

- Recursive forecast (the 2026 forecast proper). The model starts from the 24 hours before the origin, predicts the next hour, appends that prediction (with the known calendar features) to its input window, drops the oldest hour and repeats for all 8,760 hours. Only the starting window is real data, so errors accumulate and, beyond the first days, the forecast can only carry the seasonal and diurnal pattern the model has learned. It is compared with three baselines that use the same information: hour-of-day by month climatology, the same hour 52 weeks earlier, and the last value before the origin.
- Rolling one-step evaluation (the task the network was trained for). For each measured horizon hour with a valid window, the frozen model predicts it from the measured previous 24 hours. This needs measurements up to the previous hour, so it exists only where the data exist. It is compared with persistence (the last observed value) on the same rows.

Predicted concentrations are clipped at zero, both before being fed back and on output; a pollutant concentration cannot be negative. Every horizon hour is exported. Each row records whether a measurement exists (observed), the station file has the hour but no valid value (missing_value), or the file has no row for the hour (no_row_in_file). Ground truth is never imputed or interpolated.

8.3 Training and target design

Table 7. Training configuration (same for all stations and modes; no hyperparameter search).

Setting	Value
Loss	Masked MSE on standardised targets (missing targets excluded)
Optimiser	Adam [8], learning rate 0.001
Batch size / maximum epochs	256 / 60
Early stopping	validation loss, patience 6, best weights
Random seed	42 (Python, NumPy, TensorFlow)
Software	Python 3.12.14, TensorFlow 2.19.1, scikit-learn 1.9.1, pandas 3.0.6, NumPy 2.1.3

Two parameterisations of the same network are trained per station: level (predict the standardised concentration) and change (predict the difference from the last observed value). The design for the recursive forecast is chosen by a recursive forecast of the validation half-year. The design for the one-step evaluation is chosen by one-step validation loss. Neither choice sees the horizon.

Table 8. 2025 backtest: model selection on validation data (MSE, standardised units).

Station	1-step val: persistence	1-step val: level	1-step val: change	Recursive val: level	Recursive val: change	Recursive design	One-step design
MIDC Ambad	0.1945	0.2371	0.1774	1.267	2.198	level	change

Gangapur Road	0.3204	0.2693	0.2601	1.502	18.95	level	change
Pandav Nagari	0.216	0.2312	0.1756	2.327	15.45	level	change
Hirwadi	0.2102	0.2765	0.1587	0.9208	1.398e+05	level	change

Table 9. 2026 forecast: model selection on validation data (MSE, standardised units).

Station	1-step val: persistence	1-step val: level	1-step val: change	Recursive val: level	Recursive val: change	Recursive design	One-step design
MIDC Ambad	0.1625	0.1817	0.1387	1.903	2.61	level	change
Gangapur Road	0.1109	0.1293	0.09261	8.857	5.322	change	change
Pandav Nagari	0.1218	0.1185	0.09649	2.873	1.761	change	change
Hirwadi	0.3727	0.412	0.3065	1.692	24.36	level	change

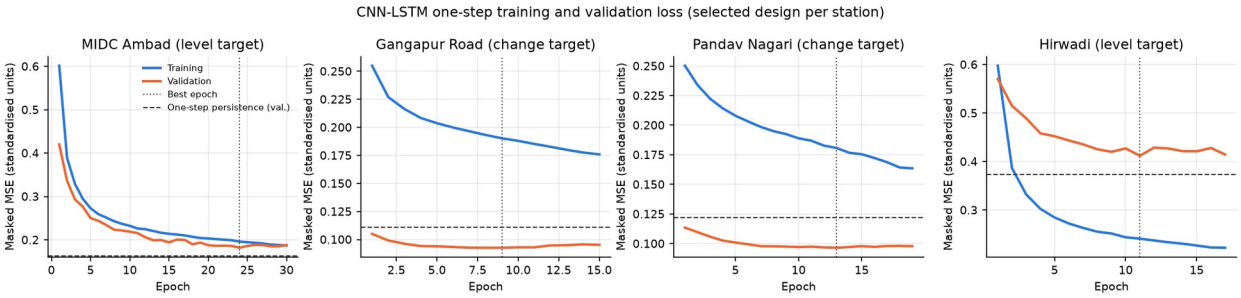


Figure 4. 2026-forecast mode: training and validation loss of the recursively selected design per station.

9. Chronological Splits

Random splitting would score the model on near-copies of its training hours and let it learn from the period it is tested on [10]. Every split here is chronological, and windows are assigned by the timestamp of the predicted hour.

Table 10. Splits per mode and station. Test = the horizon rows present in the file.

Mode	Station	Split	Start	End	Hourly rows	PM2.5 observed
2025 backtest	MIDC Ambad	train	2023-06-14 14:00:00	2024-06-30 23:00:00	9178	8371
2025 backtest	MIDC Ambad	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	4054
2025 backtest	MIDC Ambad	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	8310
2025 backtest	Gangapur Road	train	2017-01-01 00:00:00	2024-06-30 23:00:00	62760	54189
2025 backtest	Gangapur Road	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	4106
2025 backtest	Gangapur Road	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	8317
2025 backtest	Pandav Nagari	train	2023-05-22 17:00:00	2024-06-30 23:00:00	9727	9439
2025 backtest	Pandav Nagari	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	4251
2025 backtest	Pandav Nagari	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	8277
2025 backtest	Hirwadi	train	2023-05-23 16:00:00	2024-06-30 23:00:00	9704	8308
2025 backtest	Hirwadi	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	4291
2025 backtest	Hirwadi	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	8506

2026 forecast	MIDC Ambad	train	2023-06-14 14:00:00	2025-06-30 23:00:00	17938	16560
2026 forecast	MIDC Ambad	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	4175
2026 forecast	MIDC Ambad	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	4708
2026 forecast	Gangapur Road	train	2017-01-01 00:00:00	2025-06-30 23:00:00	71520	62415
2026 forecast	Gangapur Road	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	4197
2026 forecast	Gangapur Road	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	4876
2026 forecast	Pandav Nagari	train	2023-05-22 17:00:00	2025-06-30 23:00:00	18487	17784
2026 forecast	Pandav Nagari	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	4183
2026 forecast	Pandav Nagari	test	-	-	0	0
2026 forecast	Hirwadi	train	2023-05-23 16:00:00	2025-06-30 23:00:00	18464	16799
2026 forecast	Hirwadi	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	4306
2026 forecast	Hirwadi	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	4769

10. Evaluation Metrics

For n measured hours with values y_i, predictions $\hat{y}_i$ and measured mean $\bar{y}$:

$$MSE = (1/n)\,\Sigma\,(y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE}$$

$$MAE = (1/n)\,\Sigma\,|y_i - \hat{y}_i|$$

$$R^2 = 1 - \Sigma\,(y_i - \hat{y}_i)^2 / \Sigma\,(y_i - \bar{y})^2$$

- All metrics are in original units, after inverse scaling, on measured hours only. Every method is scored on exactly the same hours.
- n_valid is the number of measured hours scored. Coverage is n_valid as a share of the 8,760 horizon hours.
- R^2 is reported as undefined with fewer than two hours or zero variance, and flagged unreliable with fewer than 168 hours (one week).
- Recursive skill = 1 − RMSE_CNN-LSTM / RMSE_climatology; one-step skill = 1 − RMSE_CNN-LSTM / RMSE_persistence. Positive means the CNN-LSTM is better.
- A series whose measured horizon values take fewer than 100 distinct values is quality-flagged (stuck or quantised sensor). It is reported but left out of the pooled per-pollutant summary. Pollutants are never pooled together.

11. Results: 2025 Backtest

Models trained on data up to 31 December 2024 forecast every hour of 2025, which is measured at all four stations. This is the fair test of how the procedure behaves on a year it has never seen.

11.1 Recursive year-ahead forecast

*Table 11. 2025: recursive forecast per station and pollutant, measured hours only (µg/m³; CO in mg/m³). * = quality-flagged series; '-' = no measured hours.*

Station	Pollutant	n	Cov. %	RMSE	MAE	R^2	RMSE clim.	RMSE last yr	RMSE persist.	Skill vs clim.	R^2 flag
MIDC Ambad	PM2.5	8310	94.86	21	14.96	0.12	20.92	26.46	24.06	-0.004173	ok
MIDC Ambad	PM10	8327	95.06	60.01	43.75	-0.2961	51.64	63.78	53.07	-0.1622	ok
MIDC Ambad	NO2	8267	94.37	18.72	13.14	-1.122	11.83	15.85	12.97	-0.5826	ok
MIDC Ambad	SO2	8323	95.01	8.171	5.116	0.092	8.277	10.04	8.876	0.01286	ok
MIDC Ambad	CO	8365	95.49	0.2786	0.2014	-0.009186	0.361	0.5751	0.4213	0.2281	ok
MIDC Ambad	O3	8415	96.06	20.03	14.52	0.2805	19.23	23.54	24.11	-0.04166	ok
MIDC Ambad	NH3	8222	93.86	13.17	9.957	-0.9839	9.606	10.73	12.06	-0.3706	ok
Gangapur Road	PM2.5	8317	94.94	26.92	17.72	0.03865	22.13	36.05	49.31	-0.2165	ok
Gangapur Road	PM10	8276	94.47	54.35	34.56	-0.005868	43.94	60.57	76.48	-0.237	ok
Gangapur Road	NO2	8024	91.6	10.46	7.329	-0.5633	14.74	12.83	11.06	0.2901	ok
Gangapur Road	SO2	8438	96.32	7.23	3.672	-0.1006	6.758	8.121	8.268	-0.06987	ok
Gangapur Road	CO	8510	97.15	0.554	0.4219	-1.25	0.4205	0.7799	1.354	-0.3175	ok
Gangapur Road	O3	8486	96.87	24.72	19.64	-0.6401	23.5	22.84	26.95	-0.05172	ok
Gangapur Road	NH3	8023	91.59	14.92	13.28	-3.443	7.837	12.29	7.733	-0.9044	ok
Pandav Nagari	PM2.5	8277	94.49	31.1	16.62	-0.3888	21.19	30.79	26.42	-0.4676	ok
Pandav Nagari	PM10	8309	94.85	59.27	42.2	-0.7054	36.11	53.96	46.26	-0.6412	ok
Pandav Nagari	NO2	8347	95.29	39.01	31.38	-19.96	14.2	19.26	8.618	-1.746	ok
Pandav Nagari	SO2	8325	95.03	21.25	16.65	-3.899	9.115	11.78	9.674	-1.331	ok
Pandav Nagari	CO	8414	96.05	0.5677	0.4244	-2.16	0.257	0.3245	0.3195	-1.209	ok
Pandav Nagari	O3	8350	95.32	16.2	12.21	0.4114	16.3	21.44	42.69	0.006414	ok
Pandav Nagari	NH3	8362	95.46	33.57	26.61	-18.51	9.443	11.98	7.925	-2.555	ok
Hirwadi	PM2.5	8506	97.1	19.64	13.68	0.3125	19.27	25.78	27.81	-0.0194	ok
Hirwadi	PM10	8494	96.96	49.66	34.34	-0.01862	44.96	61.58	49.38	-0.1046	ok
Hirwadi	NO2	8503	97.07	12.12	7.785	-1.783	11.41	13.77	7.868	-0.06281	ok
Hirwadi	SO2	8171	93.28	10.8	5.21	-0.03411	10.96	11.72	12.59	0.01469	ok
Hirwadi	CO	8530	97.37	0.3733	0.2771	-0.2216	0.2973	0.3598	0.3378	-0.2556	ok
Hirwadi	O3	8528	97.35	20.95	16.48	0.1515	15.73	21.27	22.77	-0.3319	ok
Hirwadi	NH3	8472	96.71	9.822	5.781	-0.2558	10.9	10.48	9.079	0.09854	ok

Table 12. 2025: recursive forecast, unflagged stations pooled per pollutant.

Pollutant	n	MSE	RMSE	MAE	R^2	RMSE clim.	Skill vs clim.	Stations
PM2.5	33410	628	25.06	15.73	0.002715	20.89	-0.1994	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
PM10	33406	3,130	55.95	38.7	-0.1874	44.51	-0.257	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NO2	33141	534.9	23.13	14.95	-4.632	13.1	-0.765	MIDC Ambad, Gangapur Road, Pandav Nagari,

								Hirwadi
SO2	33257	171.7	13.1	7.66	-1.086	8.892	-0.4736	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
CO	33819	0.2118	0.4602	0.3314	-0.9019	0.3398	-0.3543	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
O3	33779	429.1	20.71	15.73	0.104	18.95	-0.09321	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NH3	33079	406.6	20.17	13.9	-4.859	9.528	-1.116	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi

Pooled RMSE skill against climatology ranges from -111.6% (NH3) to -9.3% (O3): the CNN-LSTM beats climatology for 0 of 7 pollutants. A year-ahead recursive forecast cannot know the weather of a given day, so after the first hours its value lies in reproducing the seasonal and daily cycle, and climatology is the baseline it has to match. It does not. Feeding predictions back lets the network settle on its own level, which can be far from the measurements for months: at Pandav Nagari, NO2, SO2 and NH3 stay at several times the measured concentrations through January-April and again from November. That station was trained on mid-2023 to mid-2024, when its NO2 was much higher than in 2025.

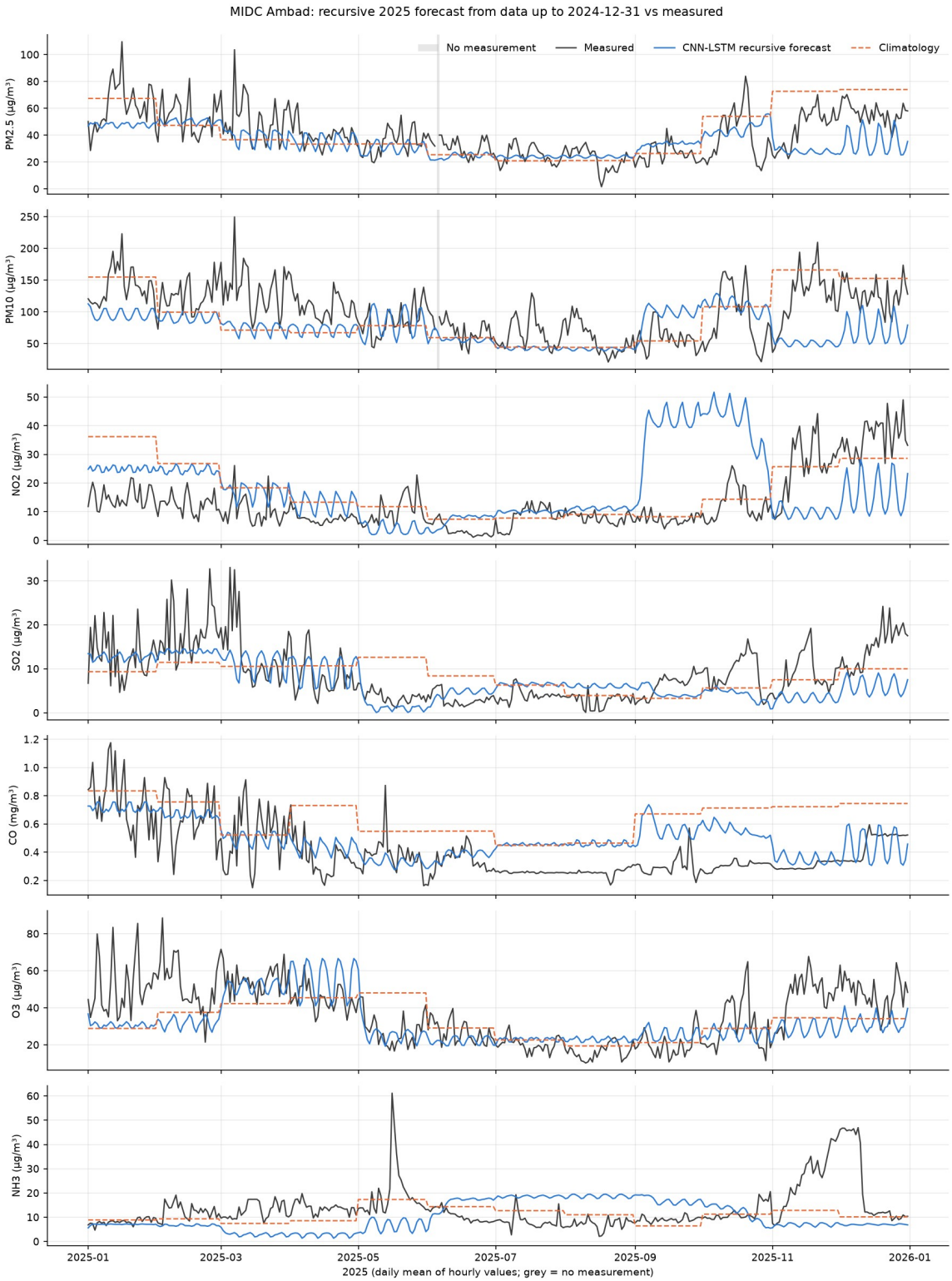


Figure 5. MIDC Ambad: 2025 recursive forecast (daily means) against measurements.

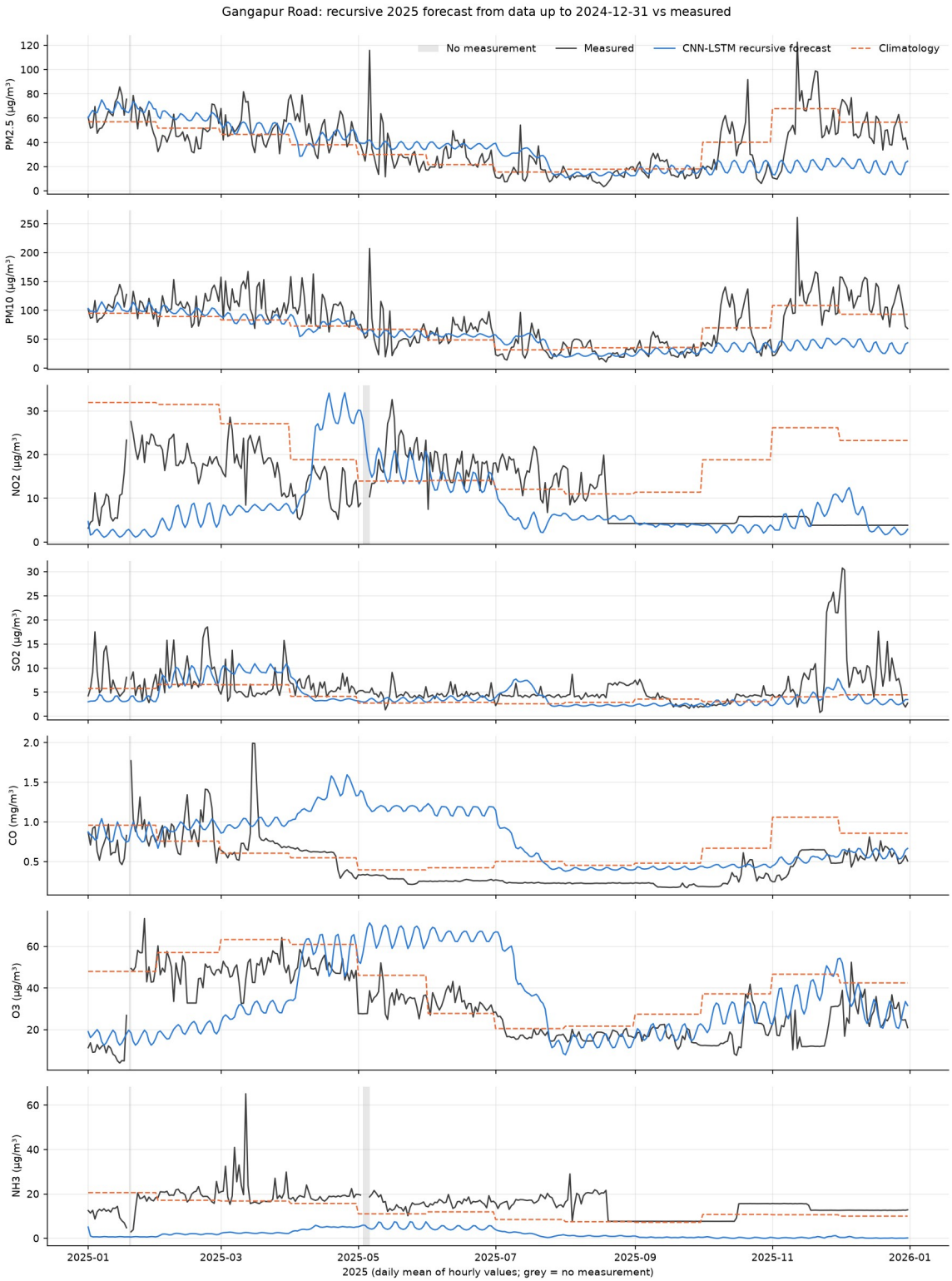


Figure 6. Gangapur Road: 2025 recursive forecast (daily means) against measurements.

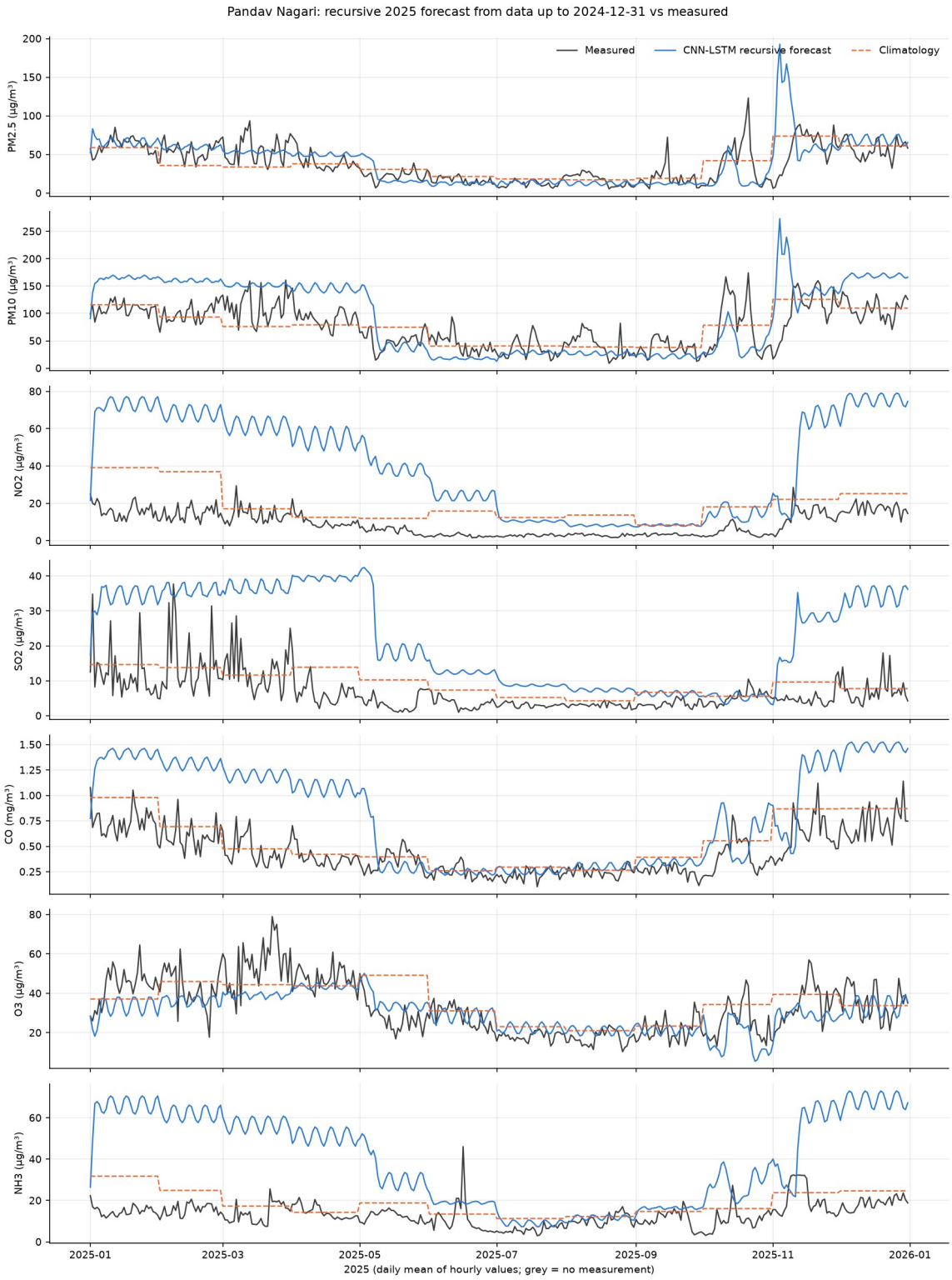


Figure 7. Pandav Nagari: 2025 recursive forecast (daily means) against measurements.

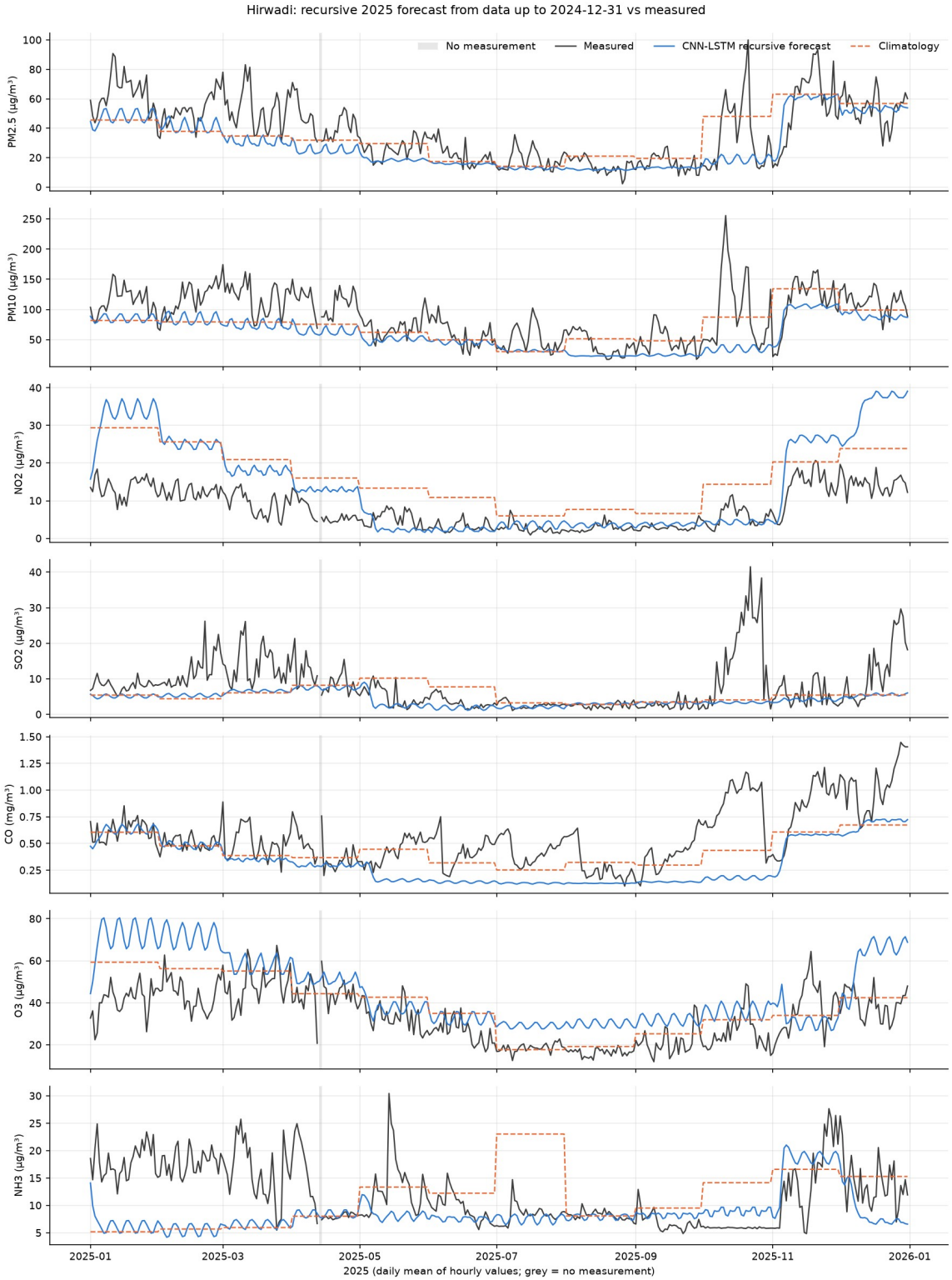


Figure 8. Hirwadi: 2025 recursive forecast (daily means) against measurements.

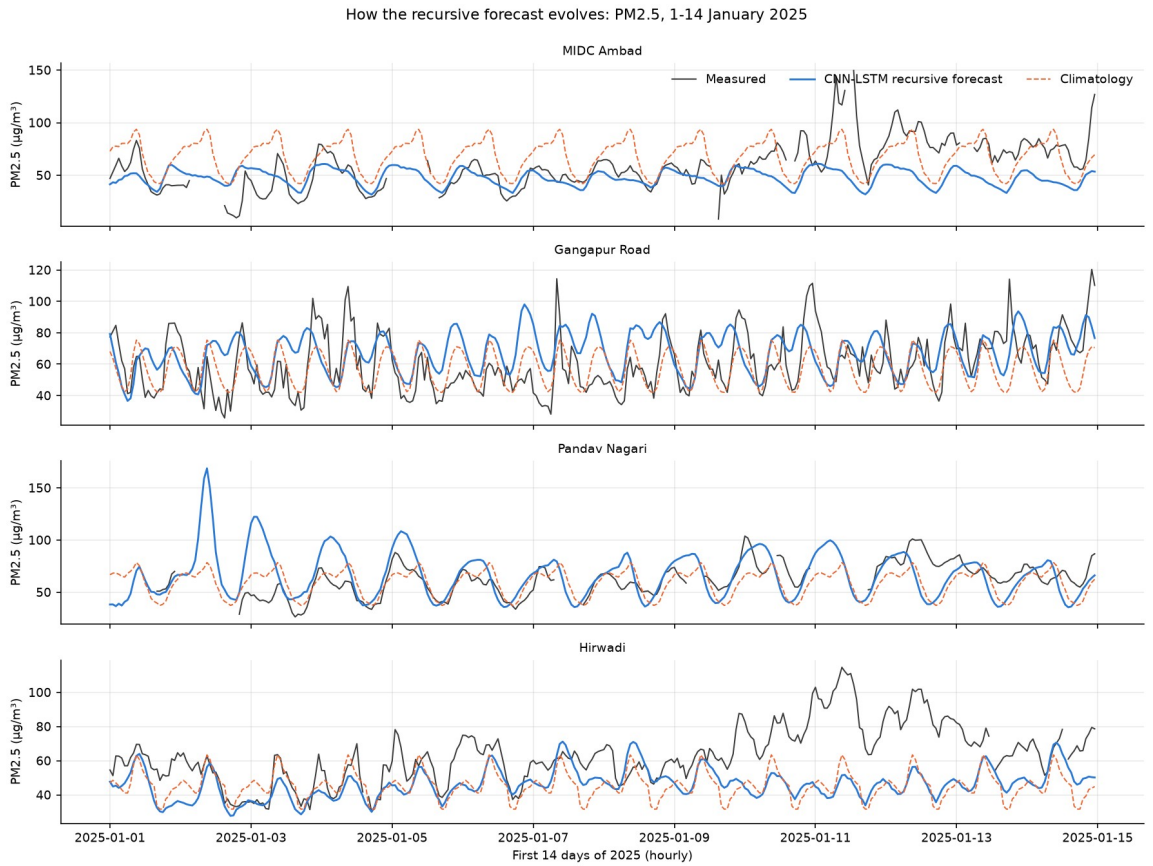


Figure 9. PM2.5 in the first 14 days of 2025: the recursive forecast drifts from the measurements within days.

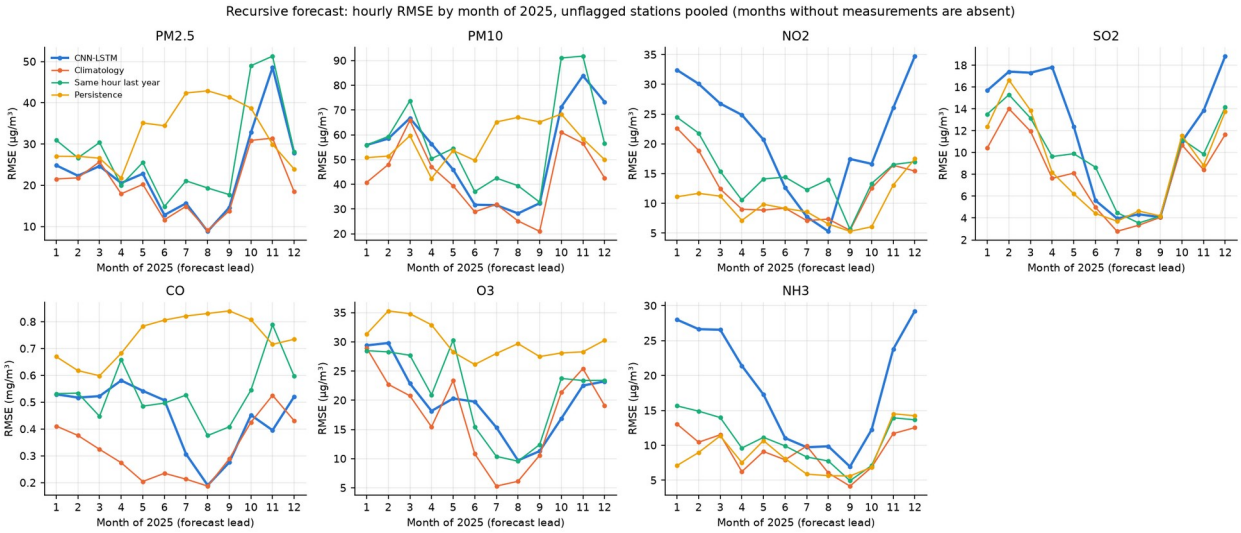


Figure 10. Recursive forecast RMSE by month of 2025 (forecast lead), unflagged stations pooled.

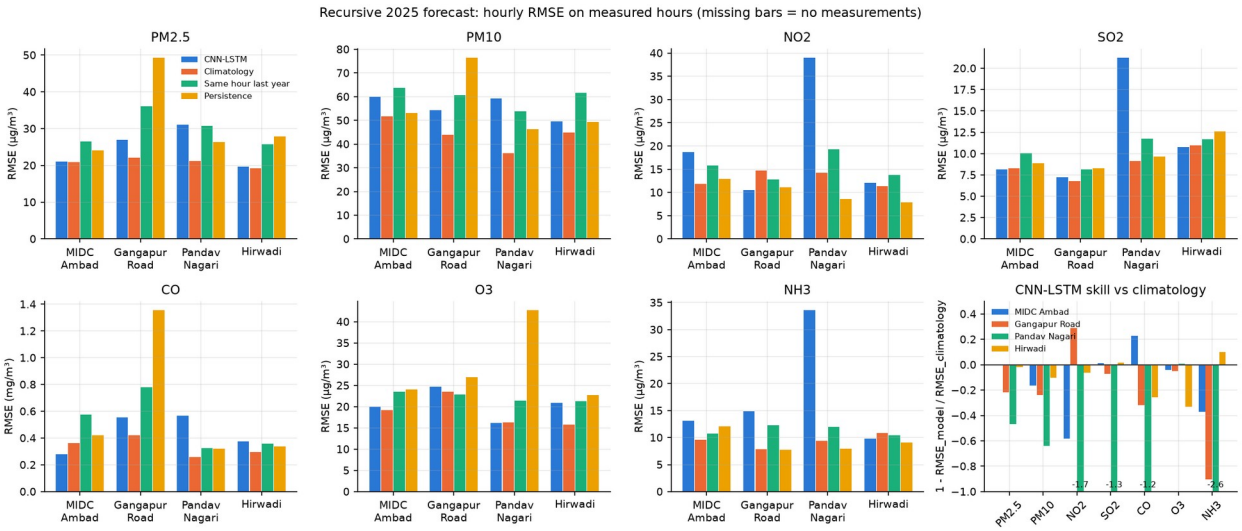


Figure 11. 2025 recursive RMSE per station and baseline, and RMSE skill against climatology.

11.2 Rolling one-step evaluation

*Table 13. 2025: rolling one-step predictions per station and pollutant. * = quality-flagged series; '-' = no measured hours.*

Station	Pollutant	n	MSE	RMSE	MAE	R^2	RMSE persist.	Skill vs persist.	R^2 flag
MIDC Ambad	PM2.5	8231	78.87	8.881	5.055	0.8434	9.28	0.04301	ok
MIDC Ambad	PM10	8248	578.5	24.05	11.27	0.7922	25.57	0.05947	ok
MIDC Ambad	NO2	8196	51.28	7.161	4.115	0.6864	6.565	-0.09073	ok
MIDC Ambad	SO2	8245	25.34	5.034	1.913	0.6555	5.367	0.06204	ok
MIDC Ambad	CO	8291	0.03664	0.1914	0.1051	0.5242	0.1819	-0.05231	ok
MIDC Ambad	O3	8338	80.57	8.976	5.826	0.8548	9.702	0.07482	ok
MIDC Ambad	NH3	8152	6.788	2.605	1.255	0.9225	2.618	0.004666	ok
Gangapur Road	PM2.5	8265	231.1	15.2	5.497	0.6911	16.03	0.05173	ok
Gangapur Road	PM10	8231	663.2	25.75	12.39	0.7729	28.13	0.08452	ok
Gangapur Road	NO2	7974	26.82	5.179	3.349	0.615	4.825	-0.07326	ok
Gangapur Road	SO2	8368	22.73	4.768	2.093	0.5234	5.718	0.1662	ok
Gangapur Road	CO	8440	0.03181	0.1784	0.08142	0.7632	0.1894	0.05823	ok
Gangapur Road	O3	8416	56.02	7.485	3.861	0.85	6.918	-0.0819	ok
Gangapur Road	NH3	7978	23.94	4.893	2.021	0.5134	4.839	-0.01105	ok
Pandav Nagari	PM2.5	8135	93.15	9.652	3.762	0.8659	10.26	0.05936	ok
Pandav Nagari	PM10	8161	220.7	14.86	7.161	0.8919	16.38	0.09281	ok
Pandav Nagari	NO2	8199	15.57	3.945	2.081	0.7819	4.29	0.08038	ok
Pandav Nagari	SO2	8177	34.22	5.85	1.999	0.6176	6.486	0.09812	ok
Pandav Nagari	CO	8266	0.01912	0.1383	0.07446	0.8104	0.1584	0.1274	ok
Pandav Nagari	O3	8202	49.97	7.069	4.604	0.8857	8.28	0.1462	ok
Pandav Nagari	NH3	8216	22.02	4.692	2.044	0.6167	4.915	0.04524	ok
Hirwadi	PM2.5	8469	58.95	7.678	4.239	0.895	7.686	0.001077	ok
Hirwadi	PM10	8457	490.8	22.15	9.587	0.7972	24.1	0.08076	ok
Hirwadi	NO2	8466	21.25	4.609	2.885	0.5989	3.9	-0.182	ok
Hirwadi	SO2	8135	68.66	8.286	3.206	0.3935	8.614	0.03808	ok
Hirwadi	CO	8493	0.02629	0.1621	0.07367	0.7697	0.1757	0.07727	ok
Hirwadi	O3	8491	96.09	9.802	5.761	0.8142	10.96	0.1053	ok
Hirwadi	NH3	8435	40.56	6.369	2.575	0.4738	6.542	0.02642	ok

Table 14. 2025: one-step predictions, unflagged stations pooled per pollutant.

Pollutant	n	RMSE	R^2	RMSE persist.	Skill vs persist.	Stations
PM2.5	33100	10.74	0.8166	11.25	0.04561	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
PM10	33097	22.11	0.8143	23.96	0.07732	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NO2	32835	5.355	0.6956	4.993	-0.07253	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
SO2	32925	6.131	0.5406	6.656	0.07894	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
CO	33490	0.1687	0.7428	0.1768	0.04557	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
O3	33447	8.416	0.8513	9.099	0.07507	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NH3	32781	4.845	0.6611	4.947	0.02062	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi

With the measured previous 24 hours as input, pooled RMSE skill against persistence ranges from -7.3% (NO2) to 7.9% (SO2). Persistence is a strong baseline one hour ahead because consecutive hourly averages are highly correlated.

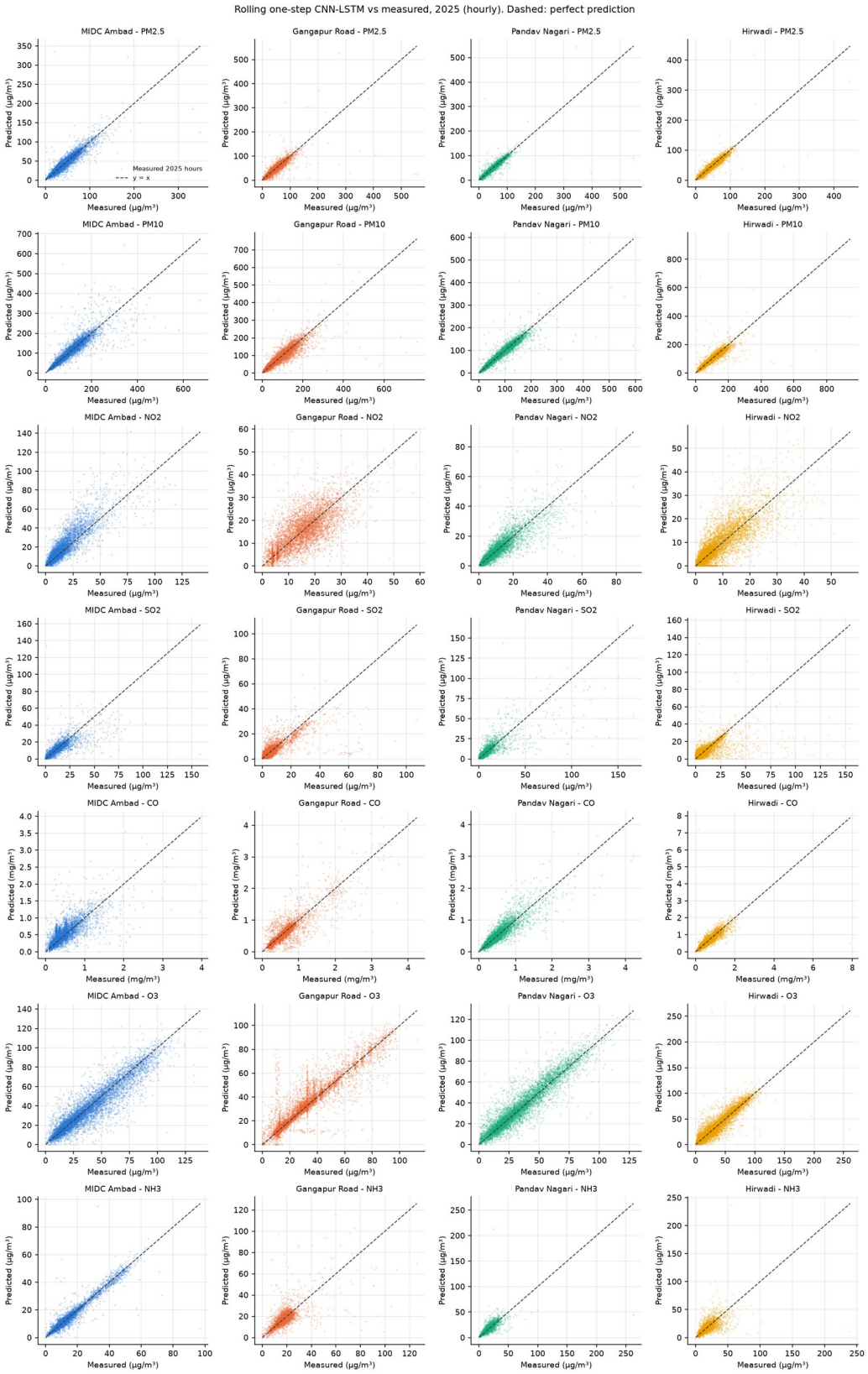


Figure 12. 2025 one-step predictions against measurements. Dashed: perfect prediction.

12. Results: 2026 Forecast

Models trained on data up to 31 December 2025 forecast every hour of 2026 at all four stations (8,760 hours each, 1 January 00:00 to 31 December 23:00). The forecast file contains every hour. Only hours labelled observed are scored: January-June and August at MIDC Ambad, Gangapur Road and Hirwadi. July and September-December 2026, and all of 2026 at Pandav Nagari, are forecasts without ground truth. Because those months are missing, the 2026 scores do not cover a full seasonal cycle (the monsoon months July-September are mostly absent).

12.1 Plausibility of the unmeasured months

Metrics cannot judge the months without measurements, so Table 15 compares the monthly mean PM2.5 forecast with the station's climatology for every month of 2026. Large ratios in either direction mark drift of the recursive forecast rather than an expected change in air quality.

Table 15. 2026 monthly mean PM2.5 (µg/m³): recursive CNN-LSTM forecast against climatology.

Station	Series	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
MIDC Ambad	Forecast	56	41	49	45	35	26	26	33	42	46	36	61
MIDC Ambad	Climatology	65	50	44	37	34	28	22	21	26	50	63	68
Gangapur Road	Forecast	82	83	73	77	83	3	0	1	16	21	112	325
Gangapur Road	Climatology	58	51	47	40	30	23	16	17	18	39	66	56
Pandav Nagari	Forecast	9	2	2	3	15	39	54	63	62	64	55	19
Pandav Nagari	Climatology	60	44	45	39	27	20	16	17	19	40	68	60
Hirwadi	Forecast	13	15	17	23	27	19	16	16	17	17	18	15
Hirwadi	Climatology	56	45	44	38	28	19	16	18	18	44	60	56

Months where the forecast is more than three times, or less than a fifth of, climatology: Gangapur Road (Jun, Jul, Aug, Dec; change design); Pandav Nagari (Jan, Feb, Mar, Apr, Jul, Aug, Sep; change design). The most extreme is Gangapur Road in December: 325 µg/m³ against a climatology of 56. With the change design, selected on validation for these stations, each step adds a predicted change to the previous prediction, and the accumulated changes carry the forecast away from any realistic level over months. These values, and the AQI derived from them, should not be used as an estimate of 2026 air quality.

12.2 Recursive forecast on measured hours

*Table 16. 2026: recursive forecast per station and pollutant, measured hours only (µg/m³; CO in mg/m³). * = quality-flagged series; '-' = no measured hours.*

Station	Pollutant	n	Cov. %	RMSE	MAE	R^2	RMSE clim.	RMSE last yr	RMSE persist.	Skill vs clim.	R^2 flag
MIDC Ambad	PM2.5	4708	53.74	15.01	11.91	0.1612	15.78	23.89	21.07	0.04886	ok
MIDC Ambad	PM10	4710	53.77	37.01	28.59	0.1667	31.89	50.19	42.71	-0.1603	ok
MIDC Ambad	NO2	4629	52.84	10.93	6.504	0.2589	8.678	12	16.36	-0.2599	ok
MIDC	SO2	4708	53.74	8.564	6.165	-0.6824	5.13	9.114	10.74	-0.6692	ok

Ambad											
MIDC Ambad	CO *	1671	19.08	0.4347	0.2956	-13.43	0.186	0.1826	0.2237	-1.337	ok
MIDC Ambad	O3	4718	53.86	16.1	13.06	0.3134	13.86	21.99	20.21	-0.1617	ok
MIDC Ambad	NH3	4654	53.13	5.591	4.697	-3.181	4.647	7.541	3.002	-0.203	ok
Gangapur Road	PM2.5	4876	55.66	44.86	37.03	-3.38	19.44	29.13	22.04	-1.308	ok
Gangapur Road	PM10	4856	55.43	52.21	39.51	-0.4798	32.53	51.12	43.41	-0.6051	ok
Gangapur Road	NO2 *	2240	25.57	15.35	13.12	-353.7	13.97	13.6	0.8257	-0.09853	ok
Gangapur Road	SO2	4329	49.42	9.865	8.019	-8.539	4.513	8.151	3.222	-1.186	ok
Gangapur Road	CO *	1753	20.01	0.6602	0.5248	-175.1	0.09962	0.2273	0.05546	-5.627	ok
Gangapur Road	O3	4987	56.93	52.95	31.59	-15.49	34.67	28.51	18.06	-0.5274	ok
Gangapur Road	NH3 *	4265	48.69	10.55	8.775	-51.82	5.089	9.518	1.476	-1.073	ok
Pandav Nagari	PM2.5	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	PM10	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	NO2	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	SO2	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	CO	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	O3	0	0	-	-	-	-	-	-	-	undefined
Pandav Nagari	NH3	0	0	-	-	-	-	-	-	-	undefined
Hirwadi	PM2.5	4769	54.44	22.83	17.21	-0.7567	12.16	20.42	23.74	-0.8785	ok
Hirwadi	PM10	4767	54.42	46.72	35.42	-0.6637	28.6	47.93	38.32	-0.6332	ok
Hirwadi	NO2	4520	51.6	14.87	10.91	-4.652	8.572	6.196	6.71	-0.7346	ok
Hirwadi	SO2	4573	52.2	14.27	8.668	-0.3562	12.54	14.38	13.94	-0.1374	ok
Hirwadi	CO	2524	28.81	0.4614	0.364	-0.6014	0.3482	0.3916	0.8516	-0.3252	ok
Hirwadi	O3	4779	54.55	19.12	13.82	0.4082	16.94	20.79	25.13	-0.1289	ok
Hirwadi	NH3	4632	52.88	13.64	10.35	-1.325	9.185	12.09	12.08	-0.4853	ok

Table 17. 2026: recursive forecast, unflagged stations pooled per pollutant.

Pollutant	n	MSE	RMSE	MAE	R^2	RMSE clim.	Skill vs clim.	Stations
PM2.5	14353	930.8	30.51	22.21	-1.692	16.1	-0.8951	MIDC Ambad, Gangapur Road, Hirwadi
PM10	14333	2,099	45.82	34.56	-0.252	31.06	-0.4751	MIDC Ambad, Gangapur Road, Hirwadi
NO2	9149	169.7	13.03	8.682	-0.5463	8.626	-0.5102	MIDC Ambad, Hirwadi
SO2	13610	124.7	11.17	7.596	-0.55	8.273	-0.3498	MIDC Ambad, Gangapur Road, Hirwadi
CO	2524	0.2129	0.4614	0.364	-0.6014	0.3482	-0.3252	Hirwadi
O3	14484	1,170	34.21	19.69	-1.726	23.9	-0.4317	MIDC Ambad, Gangapur Road, Hirwadi
NH3	9286	108.5	10.42	7.515	-1.405	7.274	-0.4321	MIDC Ambad, Hirwadi

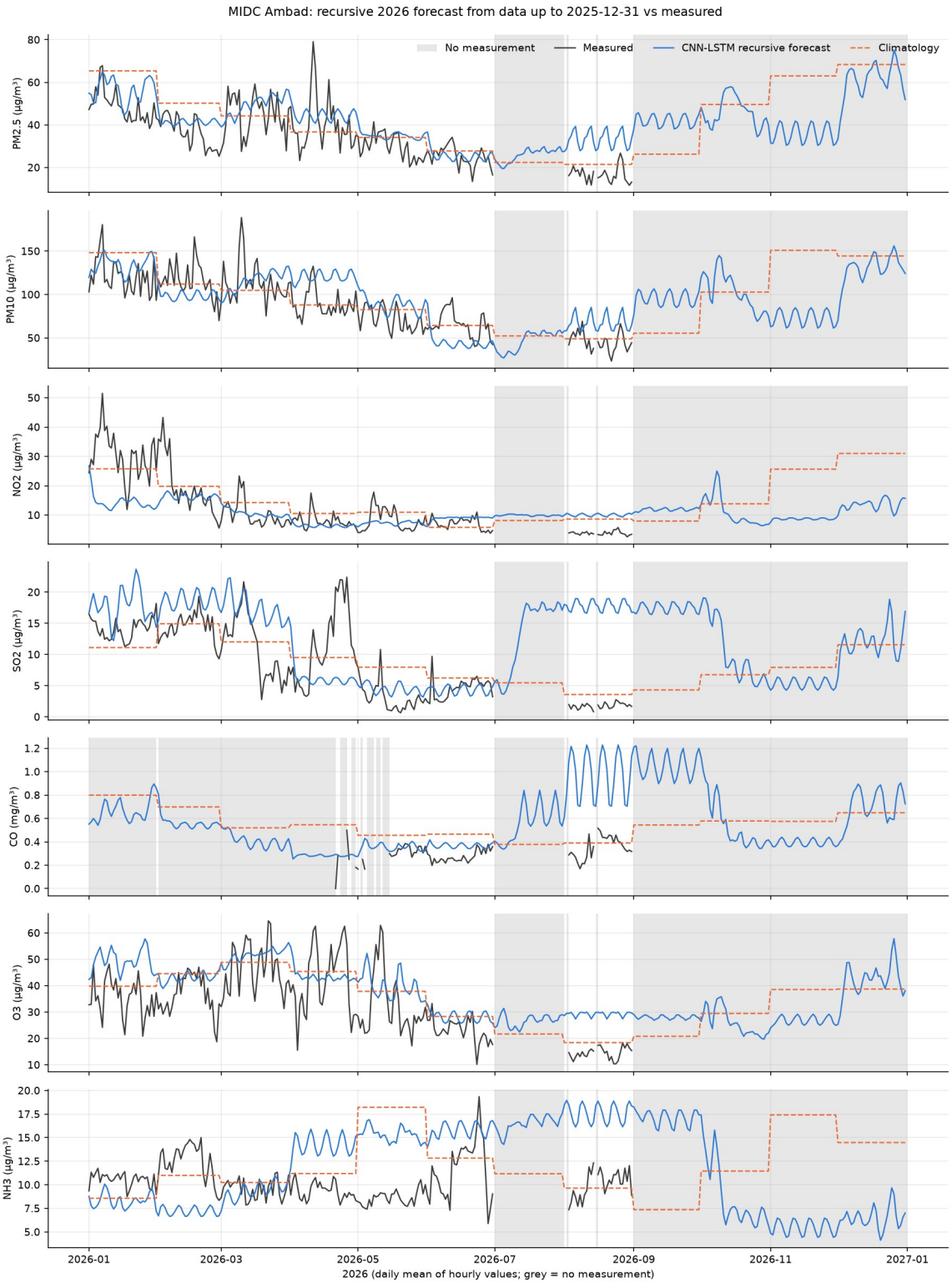


Figure 13. MIDC Ambad: 2026 recursive forecast (daily means), with measurements where they exist. Grey: no measurement.

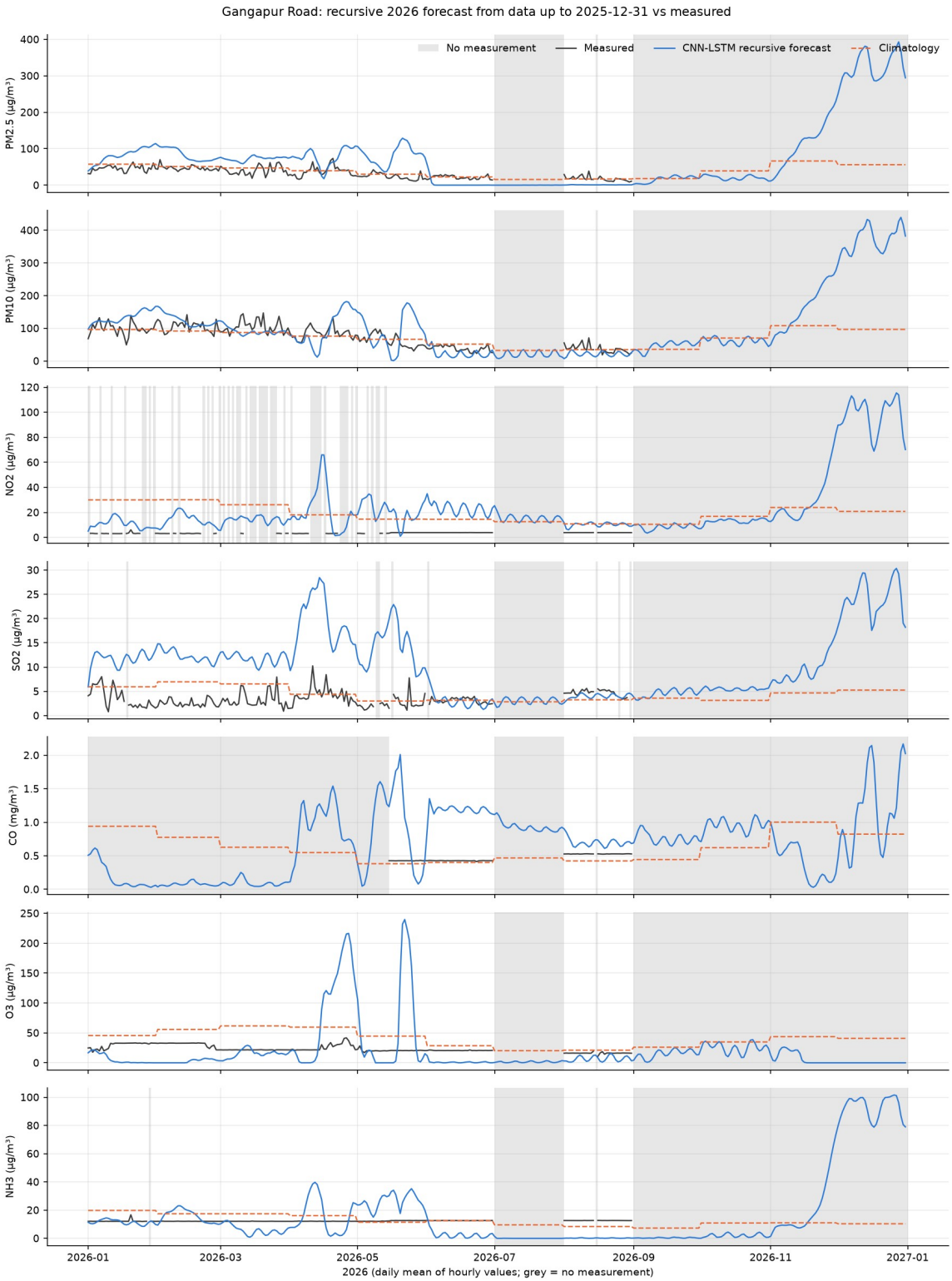


Figure 14. Gangapur Road: 2026 recursive forecast (daily means), with measurements where they exist. Grey: no measurement.

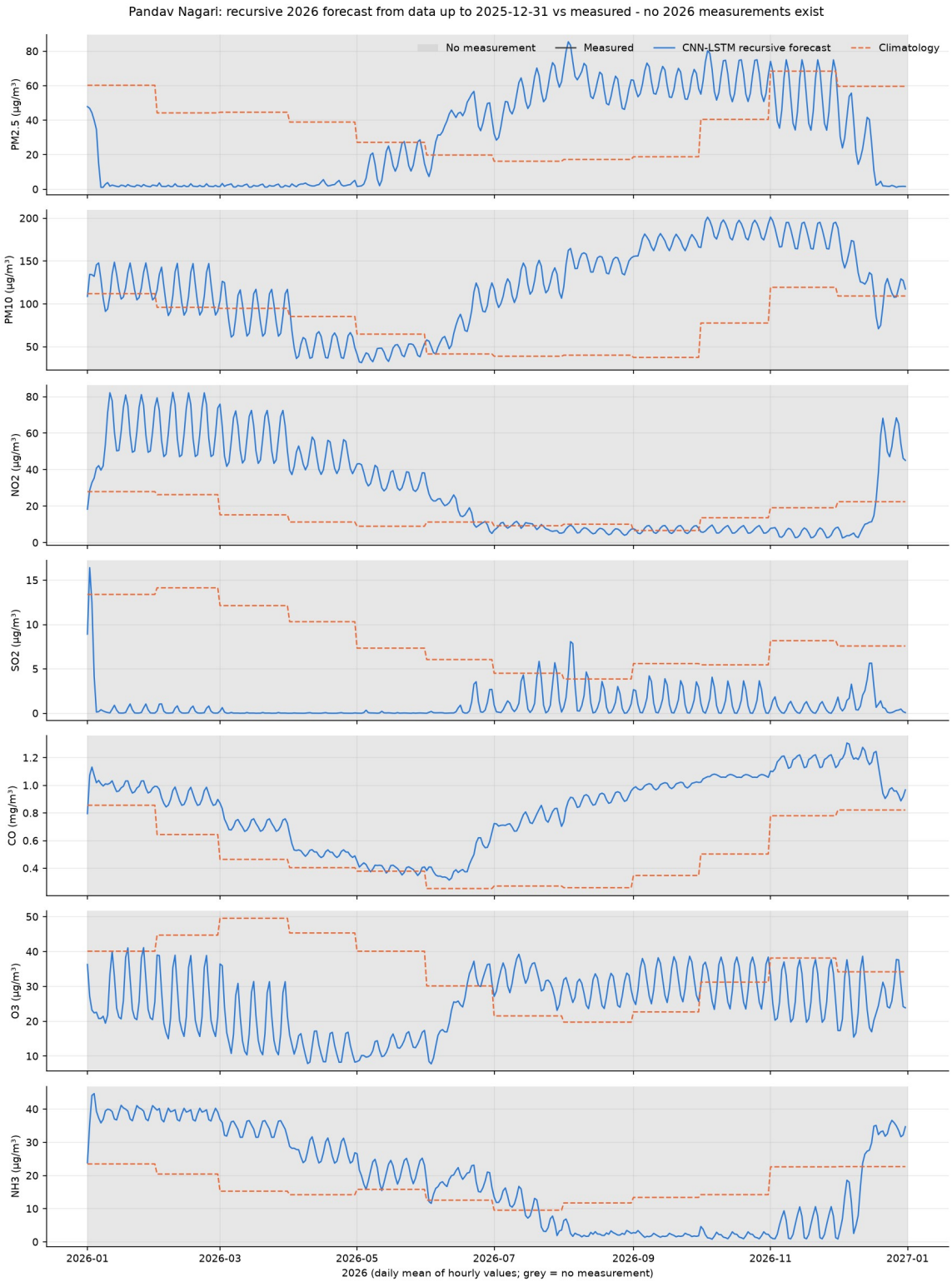


Figure 15. Pandav Nagari: 2026 recursive forecast (daily means), with measurements where they exist. Grey: no measurement.

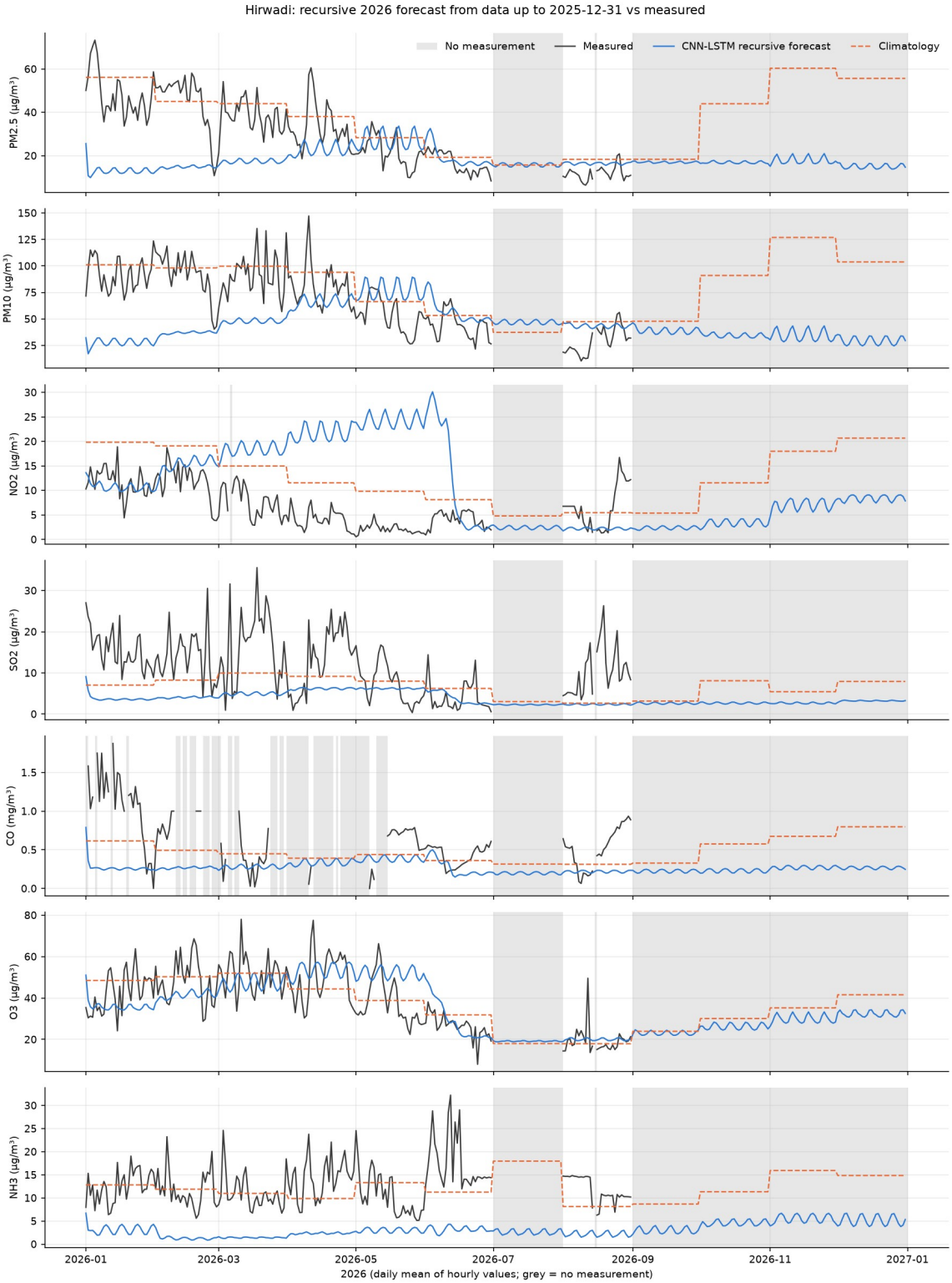


Figure 16. Hirwadi: 2026 recursive forecast (daily means), with measurements where they exist. Grey: no measurement.

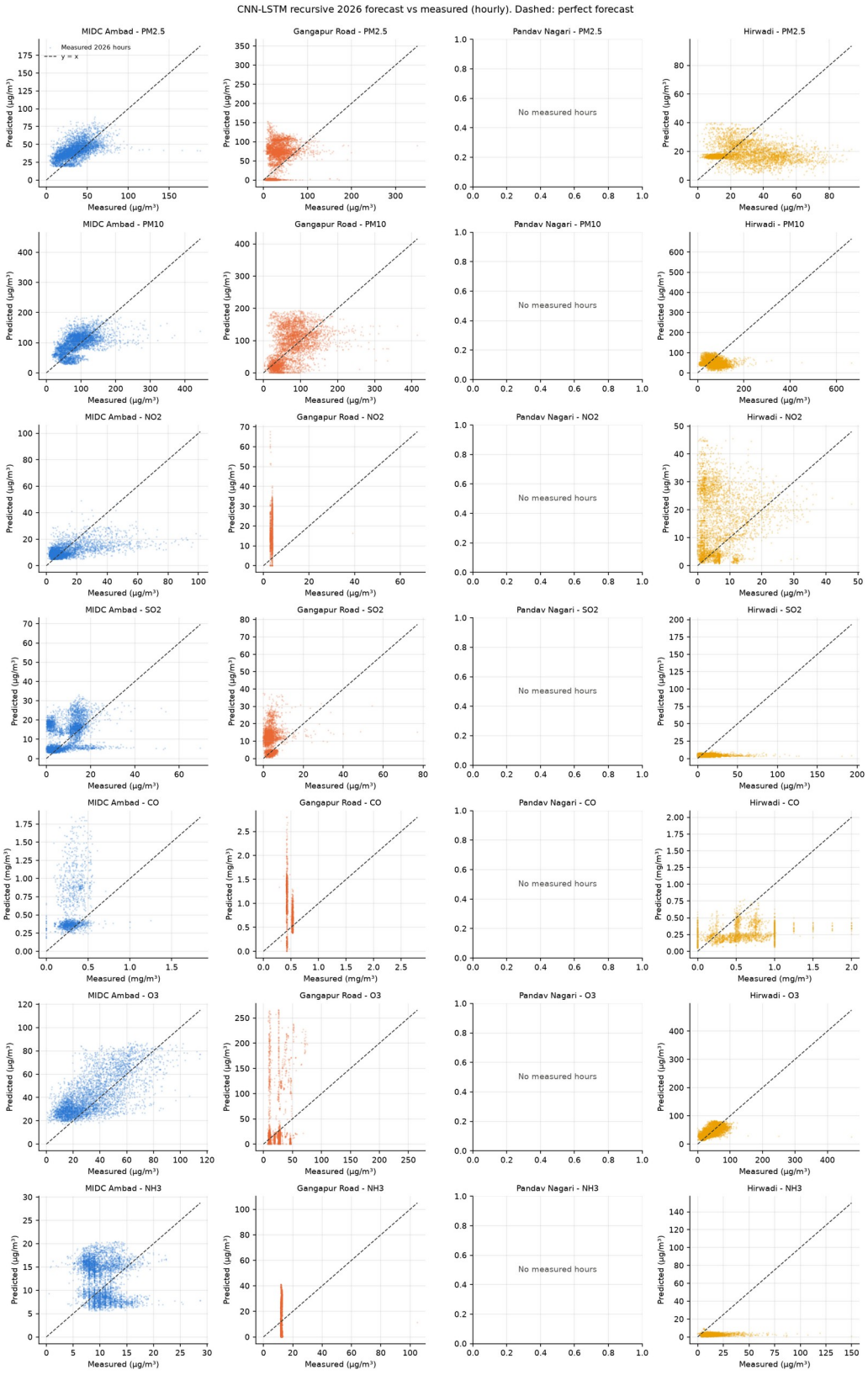


Figure 17. 2026 recursive forecast against measured hourly values.

12.3 Rolling one-step evaluation on measured hours

*Table 18. 2026: rolling one-step predictions per station and pollutant. * = quality-flagged series; '-' = no measured hours.*

Station	Pollutant	n	MSE	RMSE	MAE	R^2	RMSE persist.	Skill vs persist.	R^2 flag
MIDC Ambad	PM2.5	4542	35.71	5.976	3.981	0.8656	6.106	0.02138	ok
MIDC Ambad	PM10	4544	238.6	15.45	8.786	0.8544	17.98	0.1407	ok
MIDC Ambad	NO2	4467	25.87	5.086	2.794	0.8402	5.958	0.1463	ok
MIDC Ambad	SO2	4542	5.725	2.393	1.278	0.8637	2.496	0.04153	ok
MIDC Ambad	CO *	1574	0.007139	0.08449	0.05104	0.4433	0.08424	-0.002951	ok
MIDC Ambad	O3	4552	41.55	6.446	4.464	0.8903	7.68	0.1606	ok
MIDC Ambad	NH3	4489	1.908	1.381	0.8995	0.7476	1.378	-0.0026	ok
Gangapur Road	PM2.5	4800	135.4	11.64	5.337	0.7073	12.26	0.05065	ok
Gangapur Road	PM10	4780	350.1	18.71	10.8	0.8108	20.07	0.06767	ok
Gangapur Road	NO2 *	2190	1.426	1.194	0.4078	-1.101	1.095	-0.09071	ok
Gangapur Road	SO2	4259	7.392	2.719	1.263	0.2832	2.898	0.06186	ok
Gangapur Road	CO *	1705	0.003825	0.06184	0.05053	-0.5708	0.0119	-4.196	ok
Gangapur Road	O3	4905	53.51	7.315	3.199	0.687	7.344	0.003936	ok
Gangapur Road	NH3 *	4198	3.911	1.978	0.3034	-0.8288	2.038	0.02957	ok
Pandav Nagari	PM2.5	0	-	-	-	-	-	-	undefined
Pandav Nagari	PM10	0	-	-	-	-	-	-	undefined
Pandav Nagari	NO2	0	-	-	-	-	-	-	undefined
Pandav Nagari	SO2	0	-	-	-	-	-	-	undefined
Pandav Nagari	CO	0	-	-	-	-	-	-	undefined
Pandav Nagari	O3	0	-	-	-	-	-	-	undefined
Pandav Nagari	NH3	0	-	-	-	-	-	-	undefined
Hirwadi	PM2.5	4632	20.22	4.497	3.079	0.9311	4.789	0.06101	ok
Hirwadi	PM10	4628	305.6	17.48	7.233	0.765	18.45	0.05241	ok
Hirwadi	NO2	4402	11.66	3.414	2.091	0.7002	3.489	0.02148	ok
Hirwadi	SO2	4436	64.96	8.06	2.881	0.5709	8.695	0.07307	ok
Hirwadi	CO	2407	0.03036	0.1742	0.08154	0.773	0.1842	0.0542	ok
Hirwadi	O3	4642	159.5	12.63	6.087	0.7411	13.68	0.07715	ok
Hirwadi	NH3	4508	64.75	8.047	3.561	0.2009	8.327	0.03363	ok

Table 19. 2026: one-step predictions, unflagged stations pooled per pollutant.

Pollutant	n	RMSE	R^2	RMSE persist.	Skill vs persist.	Stations
PM2.5	13974	8.051	0.8125	8.445	0.04671	MIDC Ambad, Gangapur Road, Hirwadi
PM10	13952	17.29	0.8218	18.87	0.08368	MIDC Ambad, Gangapur Road, Hirwadi
NO2	8869	4.338	0.8291	4.891	0.1131	MIDC Ambad, Hirwadi
SO2	13237	5.11	0.6747	5.493	0.0698	MIDC Ambad, Gangapur Road, Hirwadi
CO	2407	0.1742	0.773	0.1842	0.0542	Hirwadi
O3	14099	9.195	0.8033	9.973	0.07804	MIDC Ambad, Gangapur Road, Hirwadi
NH3	8997	5.779	0.271	5.974	0.03265	MIDC Ambad, Hirwadi

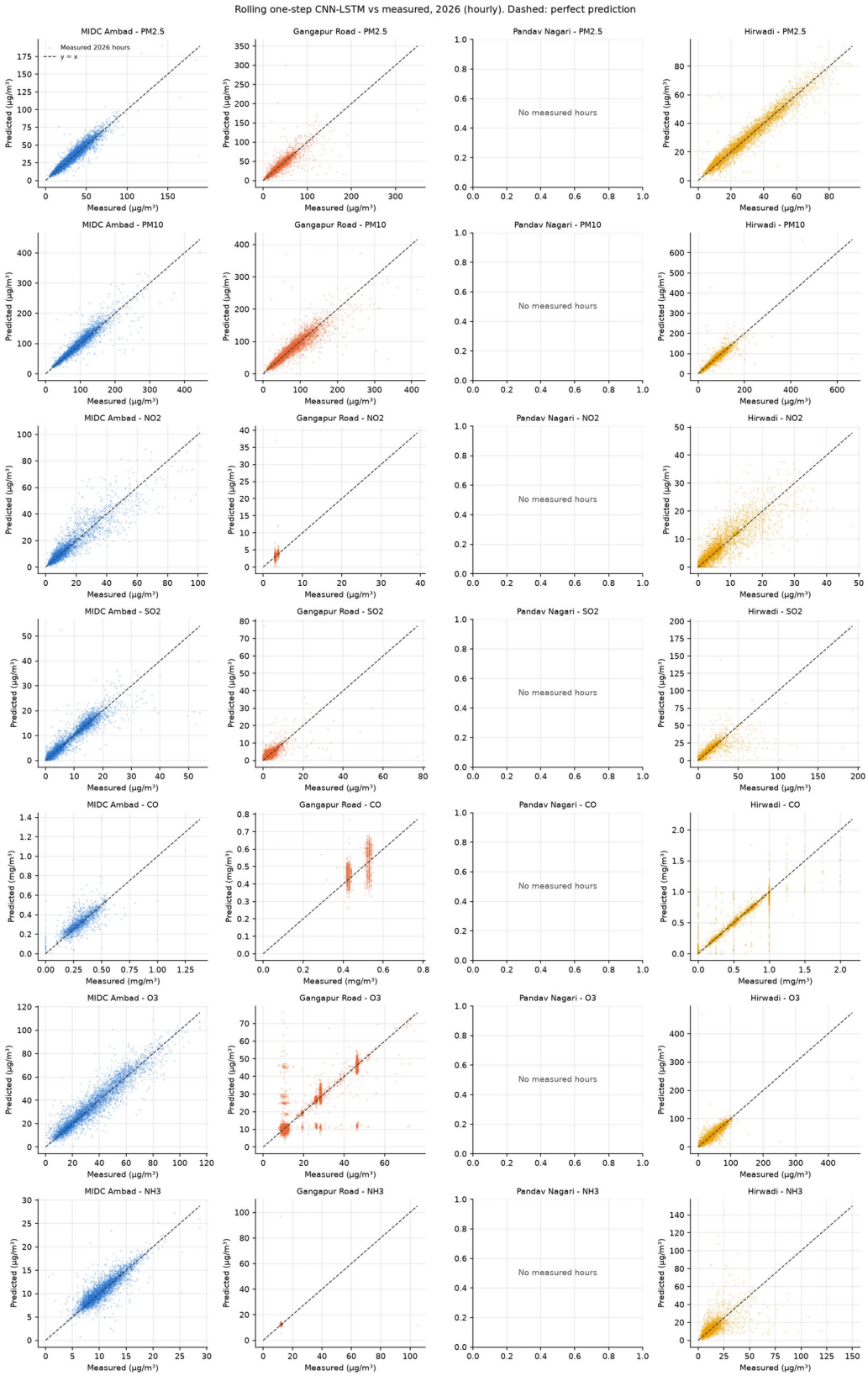


Figure 18. 2026 one-step predictions against measurements. Pandav Nagari has no 2026 measurements.

13. Predicted AQI

The model predicts pollutant concentrations; AQI is derived from them afterwards with the CPCB National AQI method [11]. Sub-indices use 24-hour means for PM2.5, PM10, NO2, SO2 and NH3 and the maximum 8-hour mean for CO and O3. The AQI is the largest sub-index and is computed only when at least three sub-indices, including PM2.5 or PM10, are available. A measured daily mean needs at least 16 measured hours. The Severe band has no published upper concentration, so it is extended linearly with the width of the Very Poor band and capped at 500. Predicted AQI is a model estimate from the recursive forecast. Measured AQI is computed from the station data with the same rules, not taken from an official bulletin.

Table 20. 2025 backtest: daily AQI from the recursive forecast against measured AQI.

Station	Forecast days	Days with measured AQI	Category agreement %	AQI RMSE	Mean predicted AQI	Mean measured AQI
MIDC Ambad	365	349	37.82	43.9	74.05	95.66
Gangapur Road	365	352	54.55	42.1	63.82	79.58
Pandav Nagari	365	347	60.23	52.89	91.36	79.27
Hirwadi	365	358	53.07	35.36	67.85	88.42

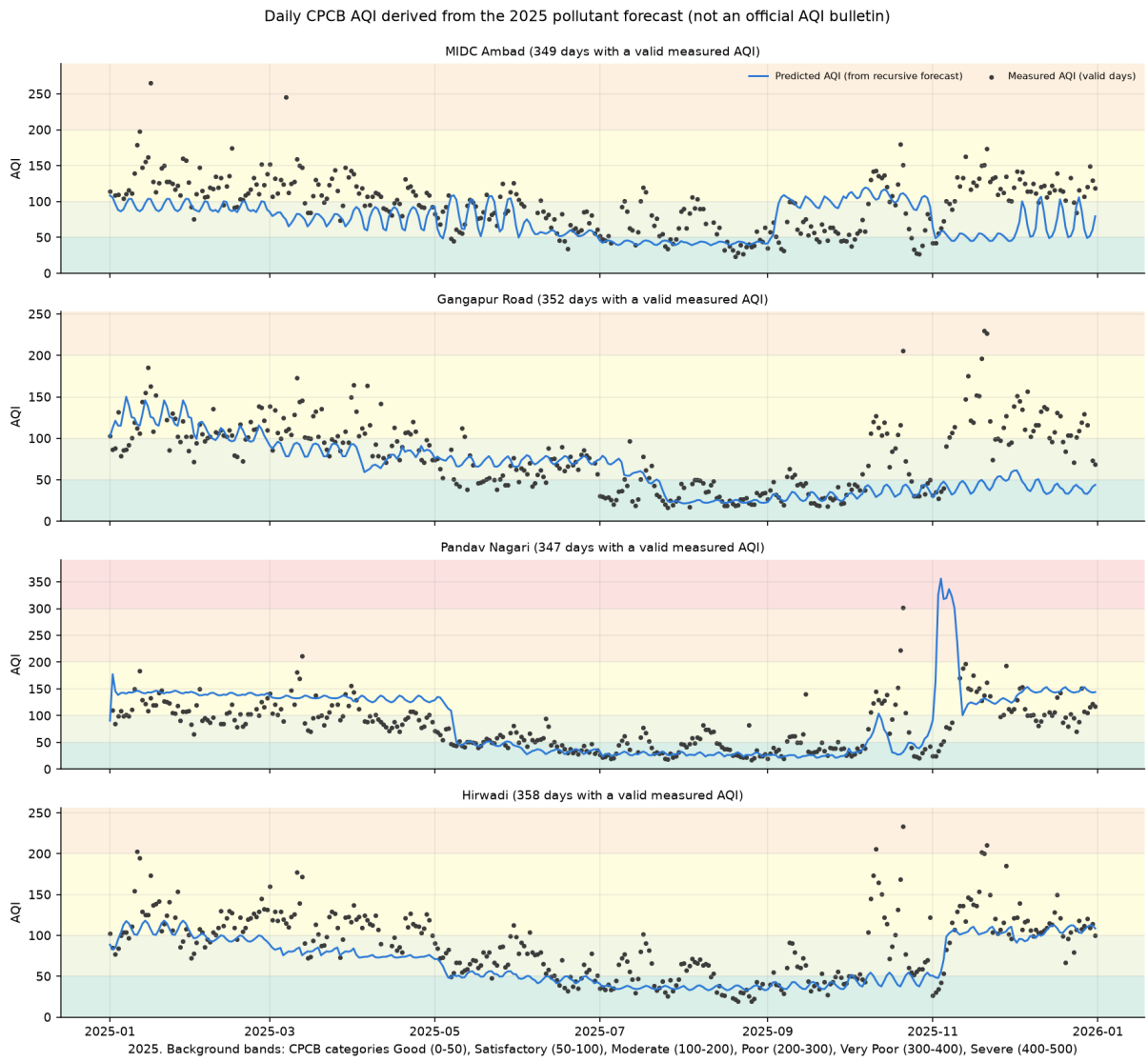


Figure 19. 2025 backtest: daily AQI derived from the forecast, with measured AQI on valid days. Bands: CPCB categories.

Table 21. 2026 forecast: daily AQI from the recursive forecast against measured AQI.

Station	Forecast days	Days with measured AQI	Category agreement %	AQI RMSE	Mean predicted AQI	Mean measured AQI
MIDC Ambad	365	196	52.04	23.79	90.45	88.26
Gangapur Road	365	208	31.73	94.17	156.1	75.61
Pandav Nagari	365	0	-	-	111.7	-
Hirwadi	365	198	66.16	30.23	52.14	73.05

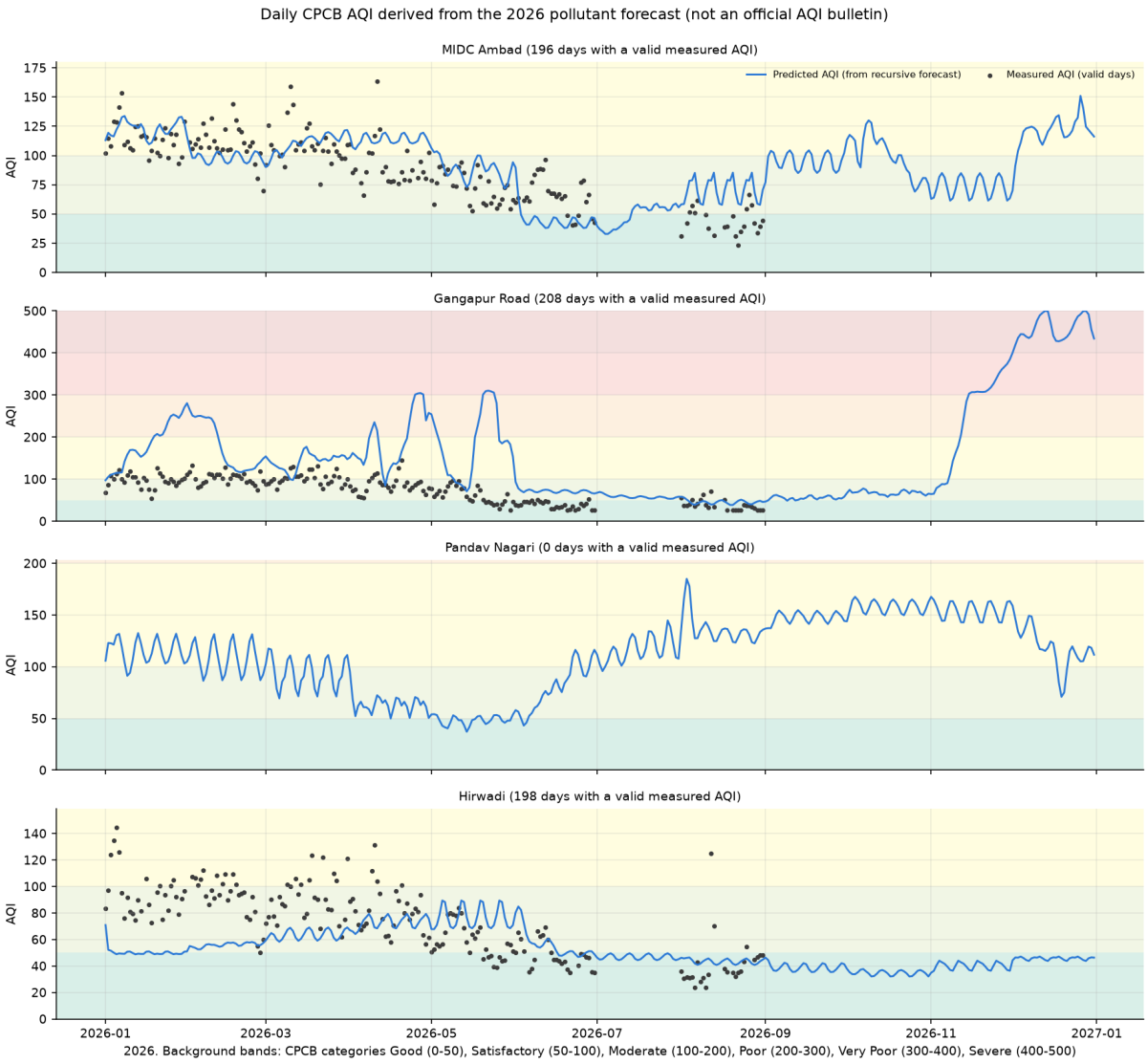


Figure 20. 2026 forecast: daily AQI derived from the forecast, with measured AQI on valid days. Bands: CPCB categories.

14. Discussion

Leakage checks. A separate script (scripts/verify_outputs.py) runs for each mode and re-derives the results from the raw CSV files and the saved models. It replays the first 48 hours of every station's recursive forecast from raw data truncated at the cutoff and reproduces the export, so the forecast does not depend on any later measurement. It rebuilds 20 random one-step input windows per station from the raw files, confirming that each prediction uses exactly the 24 preceding hours. It also checks that the scaler means equal the training-period means, that every horizon hour is present once, that missing hours carry no actual value, and that every metric matches a recomputation from the exported predictions.

Quality-flagged series. In 2026 the following series take fewer than 100 distinct measured values: MIDC Ambad CO, Gangapur Road NO2, Gangapur Road CO, Gangapur Road NH3. They are reported in every table but excluded from pooled summaries; they reflect instrument behaviour, not the model.

Recursive versus one-step. The two scores answer different questions. The one-step score says how well the network corrects persistence when fresh measurements arrive every hour. The recursive score says what can be said about a given hour months in advance, which is mostly its seasonal and diurnal expectation. The 2026 forecast file is the recursive forecast. The one-step results show what an operational next-hour system would achieve once 2026 data arrives.

15. Limitations

- 2026 ground truth covers January-June and August at three stations only; July and September-December are unmeasured, and Pandav Nagari has no 2026 data. The 2026 scores therefore miss most of the monsoon and the post-monsoon season.
- A recursive hourly forecast months ahead carries no information about specific weather events. Its hourly values should be read as an expected pattern, not as a prediction of individual hours.
- No uncertainty intervals: one training run per model (seed 42). Recursive errors accumulate with lead time, as the RMSE-by-month figure shows.
- Three stations have about two and a half years of training data (mid-2023 to the cutoff), and Gangapur Road's long history includes years whose sensor behaviour differs from 2026.
- Meteorological inputs are excluded because of missing or inconsistent sensors.
- The quality flag and the 12-hour stuck rule are heuristics; some degraded readings may remain.
- Derived AQI depends on the CPCB breakpoints as implemented here, and on the assumption for the open-ended Severe band.

16. Conclusion

The pipeline produces a leakage-free, timestamp-aligned hourly forecast of all of 2026 for all four stations, and scores it only where measurements exist. The independent verifier and the test suite confirm the horizon, the labelling of missing ground truth, the absence of post-cutoff information, and every metric.

Accuracy is a separate question, and the answer is not flattering. As a year-ahead forecast, the recursive CNN-LSTM is worse than the hour-of-day climatology for every pollutant, pooled, in both the 2025 backtest (-111.6% (NH3) to -9.3% (O3)) and the measured part of 2026 (-89.5% (PM2.5) to -32.5% (CO)). For planning-horizon estimates of 2026, the climatology column of the same forecast file is the better number. The network earns its place one hour ahead, modestly: with the previous 24 measured hours as input, its RMSE skill against persistence is -7.3% (NO2) to 7.9% (SO2) in 2025 and 3.3% (NH3) to 11.3% (NO2) in 2026.

Future work: train a direct multi-horizon model, or blend the network with climatology using weights chosen on validation, rather than feeding a one-step model its own output for 8,760 steps; add quality-

controlled meteorological inputs; report variance over several seeds; and re-score 2026 when July and September-December data and Pandav Nagari's 2026 data become available.

17. Reproducing the Results

```
NASHIK_MODE=backtest_2025 python nashik_aq_pipeline.py

NASHIK_MODE=forecast_2026 python nashik_aq_pipeline.py

python scripts/verify_outputs.py backtest_2025

python scripts/verify_outputs.py forecast_2026

python -m unittest discover tests -v

python scripts/build_report.py
```

Outputs are written to outputs/backtest_2025/ and outputs/forecast_2026/. run_metadata.json in each folder records the cutoff, horizon, features, target order, selected designs, model files, evaluation coverage, software versions and git commit (72d5052).

18. References

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